Mittuniversitetet
MID SWEDEN UNIVERSITY

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

DT168G / IU127G
Human-Computer Interaction / Interaction Design

Project group: Group 1

Obid Alrhman Aljabi
Yacoub Dawli
Sarshar Chavoshi Rad

Degree programme: Datateknik civilingenjör
Examiner: Jimmy Åhlander

October 2026

Abstract

People often listen to music, podcasts and audiobooks, but keeping track of saved and unfinished content across audio applications can be inconvenient. This project explored how these different types of audio could be organised in one interface without making navigation confusing. We developed Omnisound, a desktop-first web prototype with shared search, a Library for saved items and a Continue Listening section for unfinished audio.

We began with a project concept and paper sketches, then built an interactive first prototype (V1). We tested it with three students using three tasks: finding and saving Weezer, resuming an unfinished audiobook and browsing podcasts. We recorded task completion, time, navigation routes and participant comments. Based on the observations, we created a second prototype (V2) with a smaller Home banner, a more noticeable Continue Listening section and clearer Save/Saved controls. Three new students tested V2 using the same tasks.

All nine task attempts were completed in each round without direct help. However, only one of three V1 participants used Continue Listening directly, compared with all three V2 participants. The average time for resuming the audiobook was 27.0 seconds in V1 and 13.3 seconds in V2. The other tasks also had shorter recorded times in the second round.

The results suggest that the changes made unfinished audio easier to find for these participants. Because the tests involved small, different groups and a limited prototype, further testing is needed before drawing wider conclusions. The project shows how watching users complete real tasks can guide practical improvements to an interface.

Contents

Abstract **i**

1 Introduction **1**
1.1 Background . . . 1
1.2 Purpose . . . 2
1.3 Goals . . . 2
1.4 Delimitations . . . 3
1.5 Division of Work . . . 3

2 Theory **5**
2.1 User-Centred Design and Iteration . . . 5
2.2 Usability and User Experience . . . 5
2.3 Interface Design Principles . . . 6
2.4 Cognition and Mental Models . . . 7
2.5 User Research and Evaluation Methods . . . 7
2.6 Research Ethics . . . 8

3 Method **9**
3.1 Overall Design Process . . . 9
3.2 Initial Design Studio . . . 9
3.3 Data Collection . . . 10
3.4 Data Analysis and Requirements . . . 10
3.4.1 User Goals . . . 11
3.4.2 Usability Goals . . . 11
3.5 Prototyping Methods . . . 12
3.6 Usability Testing . . . 12
3.7 Iterative Evaluation . . . 14
3.8 Research Ethics . . . 14

4 Design **16**
4.1 Initial Design Alternatives . . . 16
4.2 Lo-Fi Sketches . . . 16
4.3 Omnisound Interface Structure . . . 18
4.4 Hi-Fi Prototype V1 . . . 19
4.5 Design Decisions and HCI Connections . . . 21

4.6 Redesign Decisions . . . 21
4.7 Hi-Fi Prototype V2 . . . 23

5 Results **24**
5.1 Data Collection Results . . . 24
5.2 Identified User Needs . . . 25
5.3 Usability Test 1 Results . . . 25
5.4 Usability Test 2 Results . . . 26
5.5 Comparison Between Iterations . . . 27

6 Discussion **29**
6.1 Interpretation of Results . . . 29
6.2 Connection Between Theory and Findings . . . 29
6.3 Strengths and Limitations . . . 30
6.4 Evaluation of Methods . . . 31
6.5 Future Work . . . 31
6.6 Conclusion . . . 32

References **33**

A Additional Prototype Screens **35**
A.1 Home Screens . . . 35
A.2 Library and Saving . . . 36

B Usability Testing Documentation **37**
B.1 V1 Usability Testing . . . 38
B.2 V2 Usability Testing . . . 57

Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Existing services already take different approaches to this organisation. Spotify combines music, podcasts and audiobooks in one service, while keeping the content types identifiable within its listening experience [1]. Apple offers separate applications for music, podcasts and audiobooks through Apple Music, Apple Podcasts and Apple Books [2]. These examples show that combining audio content is not a new idea, and neither approach is automatically easier for every user. Omnisound instead explores a specific interface question: whether shared search, a common library and clear content categories can make it easier to find saved material and continue unfinished listening without confusing the different audio types.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. It primarily targets students

who regularly listen to music, podcasts or audiobooks, including listeners who use several content types and those who mainly listen to music. The project investigates how the interface can help these listeners keep track of saved and unfinished content.

The initial high-fidelity prototype (V1) is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two main task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The first goal was to identify what students need from an application that brings music, podcasts and audiobooks together. We wanted to focus on the functions that matter most, such as finding audio, saving content for later and continuing something that was not finished. These needs would help us decide what to include in Omnisound and give us a way to evaluate the interface beyond how it looks.

Another goal was to explore different design ideas before developing the interactive prototypes. We considered different ways to organise audio categories and display content, such as separate categories or a mixed view with filters, and cards or lists. The early sketches helped us explore these ideas, while V1 and V2 showed how the design developed into an interface that users could interact with.

The final goal was to test how well people could use Omnisound. We wanted to see whether users could complete the main tasks, where they experienced difficulties and which parts of the interface needed improvement. The first usability test helped us decide what to change in V2, and the second test allowed us to compare the two versions. Together, the sketches, prototypes and usability tests helped us explain our design choices and what we learned from the project.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or subscription processing, licensing systems, backend databases or production account management. The prototype is implemented only as a front end. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

Following the first usability test, a second version of Omnisound was developed to address the problems identified during evaluation. The changes focused on reducing the size of the Home banner, making Continue Listening easier to discover, and improving the clarity of the Save/Saved control. A second usability test was then conducted using V2 and the same three core tasks, allowing the group to compare the two iterations. The testing procedures and findings are presented in the Method and Results chapters. Since Omnisound remains an interface prototype, the conclusions concern the tested interactions rather than the technical performance of a complete streaming service.

1.5 Division of Work

The project was carried out by Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad. We worked together on the early paper sketches, discussed user needs and requirements, and explored different ways to organise the interface.

Obid and Yacoub were mainly responsible for developing the two interactive prototypes, V1 and V2. The group discussed the findings from the first usability test and agreed on the changes needed for V2, including making Continue Listening easier to notice and improving the Save/Saved control.

In the initial interview plan, Yacoub was assigned to lead the interviews, while Obid and Sarshar were assigned to take notes. During the usability tests, Yacoub led the

sessions and presented the tasks to participants. Obid recorded participant behaviour and comments, while Sarshar recorded task completion times, clicks and navigation routes.

All three group members contributed to writing and reviewing the report. We worked together to explain our design decisions, present the test results and discuss what we learned throughout the project.

Chapter 2

Theory

2.1 User-Centred Design and Iteration

User-centred design begins with understanding the people who will use an interface, what they want to achieve and the circumstances in which they will use it. Rogers, Sharp and Preece explain that understanding people's activities and differences helps designers question assumptions about their needs [4, pp. 16–17]. A target group therefore provides a starting point for research, rather than a reason to assume that all its members have the same habits or expectations.

User needs and goals guide the requirements for an interface. Designers can then explore alternative ways to support those goals and represent the ideas in prototypes. Evaluation with intended users examines whether a proposed solution works as expected. A prototype is a candidate solution that can be tested by observing people using it [5, Sec. "What Is a Testable Prototype?"]. This makes evaluation part of design, rather than a check reserved for the finished product.

Iteration connects evaluation to further design decisions. Findings can lead to revised requirements, a different alternative or changes to an existing interface, followed by another evaluation [5, Sec. "Why Test a Prototype?"]. Simply producing several versions does not establish this connection. A meaningful iteration should explain which problem motivated a change and examine whether the problem remains. This principle is relevant to Omnisound's student audience and its two evaluated high-fidelity prototype iterations.

2.2 Usability and User Experience

ISO 9241-11:2018 relates usability to specified users, goals and contexts of use through effectiveness, efficiency and satisfaction. These definitions are reproduced in ISO/TR 25060:2023 [6, Sec. 3.1]. Usability concerns what people can achieve with a system in a particular situation; it cannot be established from appearance alone.

Effectiveness concerns accurate and complete achievement of a goal, while efficiency concerns the resources used in relation to that outcome [6, Secs. 3.1.7–3.1.8]. In an audio

interface, effectiveness could involve successfully saving an item and finding it again. Efficiency concerns the time and effort required to reach that outcome. Successful completion and a straightforward route are thus different aspects of the same task. Satisfaction concerns how well the experience of use meets the user's needs and expectations [6, Sec. 3.1.9].

Learnability and utility are additional useful HCI concepts. Learnability concerns how easily a person can learn to use an interface, including getting started with unfamiliar controls. Utility concerns whether the available functions support what users need to do [4, p. 21]. For example, a clearly labelled search may be easy to learn, while a way to return to unfinished audio addresses a functional need. These concepts complement the three ISO dimensions; they are not additional components of the ISO definition of usability.

User experience includes people's perceptions and responses before, during and after interaction, including anticipated use [6, Sec. 3.1.15]. Satisfaction and frustration are relevant examples. Task completion alone cannot explain how an interaction feels. Together, these concepts provide a basis for examining searching, saving, retrieving and resuming audio without treating one measure as a complete assessment of the interface.

2.3 Interface Design Principles

Visibility means making the options needed for a task apparent to the user [4, p. 28]. An available function has limited value if people cannot discover it. In an audio interface, recognisable navigation and information about listening progress can help users identify where to go and what can be continued.

Feedback communicates the outcome of an action and the current system state [7, Sec. 1]. After saving an item, a visible confirmation can help the user understand that the action succeeded. Feedback should be timely and clear enough to support the next decision, such as continuing to browse or opening the saved collection.

Consistency means using similar terms, controls and behaviour for similar interactions, both within a product and in relation to familiar conventions [7, Sec. 4]. A shared saving action across different audio types can support predictable interaction. Consistency still allows relevant differences in content, such as episode titles and audiobook chapters.

Affordances concern the actions possible in the relationship between a person and an object. Signifiers are perceivable cues that communicate what can be done [8]. In a graphical interface, a button label, icon or visual treatment can signify an available action. A control's appearance and the action it actually supports should agree. This distinction helps explain why making an action available and making it understandable are separate design concerns.

2.4 Cognition and Mental Models

A mental model is a user's understanding of how a system works. It supports predictions about what an action will do and develops through previous experience [9, Sec. "What Are Mental Models?"]. Users and designers may have different models of the same interface. Familiarity with another audio service may, for example, shape expectations about where saved items belong or whether opening unfinished content will resume it. Clear category names and navigation can help communicate how a shared audio collection is organised.

Attention concerns which information people notice while pursuing a goal. Layout and visual prominence matter because competing information can reduce the visibility of relevant content [7, Sec. 8]. A large introductory area and a route back to unfinished listening may compete for attention on the same screen. This provides a reason to examine the relative prominence of elements, without assuming that size alone determines what every user notices.

Recognition over recall means displaying options and information that users can identify, rather than requiring them to remember details [7, Sec. 6]. Visible content categories, item titles and progress information can support this recognition. For an audio collection, these cues may help someone distinguish saved material from unfinished listening and choose a suitable next action.

2.5 User Research and Evaluation Methods

Qualitative interviews explore people's experiences, needs and expectations. Open questions and follow-up questions allow participants to explain what matters to them in their own terms [10]. Interviews can inform requirements, but they collect accounts of behaviour rather than direct observations of interface use. Combining these accounts with observations can provide a fuller basis for understanding users.

Affinity diagramming organises related observations or ideas into groups that express common themes [11, Sec. "Defining Affinity Diagramming"]. It can help a design team make sense of interview notes and identify connections between needs. The grouping is an interpretation of the material, so observations that differ from a common theme should also remain visible in the analysis.

Personas are research-informed representations of users' goals, needs and expectations. Qualitative personas draw on patterns in research to help a team focus its design decisions [12]. They summarise relevant differences between users rather than reproducing every interview as a separate profile. A persona based only on assumptions should not be treated as evidence about the intended audience, and a qualitative persona does not establish how common a user type is in the wider population.

Low-fidelity prototypes, such as paper sketches, represent early ideas with limited detail. They allow designers to explore layouts and navigation before investing in a detailed interface, and can be changed or discarded readily [5, Sec. "Benefits of Low-Fidelity Prototypes"]. More detailed, interactive prototypes allow evaluation of specific

controls and task flows [5, Sec. “Benefits of High-Fidelity Prototypes”]. The appropriate level of detail depends on the question being investigated; visual polish alone does not establish that a design meets users’ needs.

Think-aloud usability testing asks participants to express their thoughts while completing representative tasks with an interface [13, Sec. “Defining Thinking Aloud Testing”]. Observations of actions, hesitation and comments can reveal difficulties and help explain why a participant chose a particular route. These observations can inform redesign, including changes to the visibility or interpretation of controls.

Qualitative usability tests also have limitations. Speaking aloud creates an unusual situation, and prompts can influence behaviour [13, Sec. “Downsides of Think-Aloud”]. Findings from a small group can identify problems in the tested interactions, but do not automatically provide statistically generalisable conclusions. Task times and navigation steps should therefore be interpreted alongside observations and the test context, rather than treated as universal measures of performance.

2.6 Research Ethics

User research should respect participants’ ability to make an informed and voluntary decision. Vetenskapsrådet’s *God forskningssed 2024* explains that people should understand what participation involves, how information will be handled and how to withdraw consent [14, pp. 62–63]. Information should be understandable to the intended participants, who should have an opportunity to ask questions and decline or stop participation without pressure.

Privacy requires responsible handling of participant information. Only information needed for the study should be collected, and it should be protected appropriately. Removing names alone may not prevent identification when other details can be combined [14, pp. 66–67]. This matters when selecting background details, quotations and observations for a report.

Accurate reporting is also an ethical responsibility. Observations and measurements should be presented honestly, with interpretations and limitations made clear [14, pp. 11–12]. Results that challenge a design should be retained rather than adjusted to support a preferred conclusion. These principles provide the basis for describing the group’s documented ethical procedures in the Method chapter.

Chapter 3

Method

3.1 Overall Design Process

The project combined early concept development, design alternatives, prototyping and two rounds of usability testing. The concept and paper sketches established the intended listening activities and possible ways to organise the interface. These ideas were developed into the first high-fidelity prototype, V1. Its evaluation provided a basis for a targeted redesign, which produced V2. A second usability test then examined the updated interface using the same core task goals. Figure 3.1 summarises this sequence.

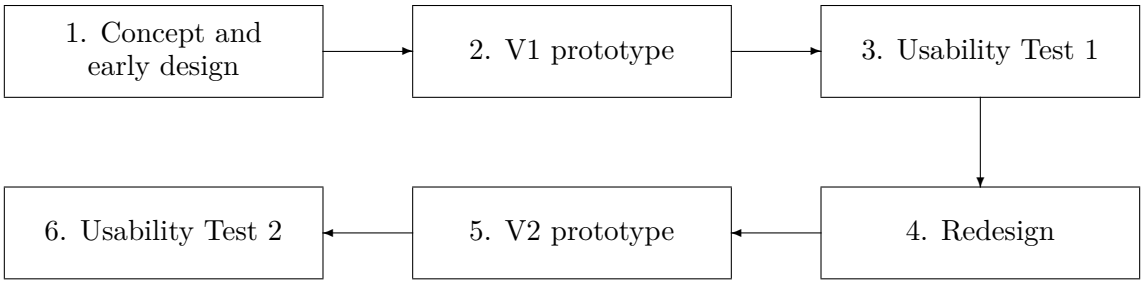


Figure 3.1: Omnisound's design and evaluation process (original project diagram).

This approach was suitable because combining audio types raises questions about how people understand categories, saved content and listening progress. An interactive prototype allowed the group to observe those decisions during use. The user-centred approach described in Section 2.1 therefore informed evaluation of concrete tasks. Iteration connected the observed problems to changes and another evaluation; creating a second version alone would not have provided that connection.

3.2 Initial Design Studio

The early design work was documented through concept material and low-fidelity sketches. The Session 4 concept sheet set out a shared audio interface for students, with finding,

saving and returning to content as its main activities. It also recorded alternative approaches to organisation: separate category tabs compared with a mixed feed and filters, and list views compared with cards. These alternatives framed the question of how to combine content while keeping its different types recognisable.

The original paper sketches explored a navigation area, grouped audio categories and a player layout. Together, they made it possible to examine where navigation, content and playback controls could fit before committing to a detailed interface. They provided concrete material for discussing interface ideas, without requiring each alternative to be fully implemented.

At this stage, the focus was the organisation of activities and information. The concept material connected possible layouts to shared search, a common library and continued listening. Keeping the drawings simple supported consideration of these relationships before visual detail and interactive behaviour became the main focus. The sketches and selected design are presented in the Design chapter.

3.3 Data Collection

To better understand what students might want from a combined audio application, the group prepared an interview plan during Project Session 2. The questions focused on which audio applications people normally used, what they liked about them, what could be improved and what they would expect from an application that brings music, podcasts and audiobooks together. The plan also included showing early interface ideas and asking about first impressions, missing features and how users would expect to complete common tasks.

Yacoub Dawli was assigned to lead the interviews, while Obid Alrhman Aljabi and Sarshar Chavoshi Rad were assigned to take handwritten notes. The questions were mostly open-ended so that participants could explain their habits and preferences in their own words. This approach was useful because the group wanted to understand what people expected from an audio interface before deciding how it should be organised [10].

During the V1 and V2 usability tests, we also asked participants about their listening habits. We wanted to know which applications they normally used, whether they listened mainly to music, podcasts or audiobooks, and which devices they preferred. Their answers gave us useful background information about the participants. Together with their comments and our observations during the tasks, this helped us understand how they experienced Omnisound.

3.4 Data Analysis and Requirements

The initial Omnisound concept and the chosen prototype tasks helped the group decide which functions were most important. The interface needed to let users find audio content, save items for later, return to their saved collection and continue listening to something

they had not finished. It was also important that users could easily tell the difference between music, podcasts and audiobooks, even though all three were available in the same application.

Rather than trying to build every feature found in a commercial audio service, the group focused on a few tasks that users could complete from beginning to end. For example, finding an artist was connected to saving that content and checking that it appeared in the Library. Resuming an audiobook required users to find unfinished content and return to the previous listening position. Browsing podcasts was another simple way to examine whether the different audio categories were easy to find. These tasks helped determine which screens, controls and interactions the prototypes needed.

The group also had to decide what could reasonably be included within the scope of a university project. Features such as subscriptions, account management and real audio streaming were left outside the prototype because the main focus was the interface and how people interacted with it. The later usability tests helped identify which parts of the interface were clear and which needed improvement. Although personas and affinity diagrams are discussed as HCI methods in Chapter 2, no completed outputs from those methods are documented in the project material, so they are not presented as steps used in the requirements analysis.

3.4.1 User Goals

The user goals describe what people should be able to do with Omnisound. Based on the project concept and the main activities selected for the prototype, we focused on four goals: finding audio content, saving something for later, returning to unfinished listening and browsing different audio categories without confusion.

For example, someone should be able to find an artist, save it and then locate it again in Library. Similarly, someone who has not finished an audiobook should be able to find it and continue from the previous position. These goals helped us decide which functions and screens to include in the prototype.

3.4.2 Usability Goals

Alongside the user goals, we wanted the main interactions to be easy to understand and complete. We focused on effectiveness, efficiency and learnability. This meant examining whether users could complete the tasks, how much time and navigation they needed, and whether they understood the interface without being shown which buttons to use.

We also wanted actions to provide clear feedback. For example, when someone saves an item, the interface should make it clear that the item has been saved and where it can be found again. During usability testing, we examined these goals through task completion, completion times, navigation routes, hesitation, participant comments and the need for assistance.

3.5 Prototyping Methods

Low-fidelity sketches and high-fidelity prototypes served different purposes. The paper drawings explored layouts and relationships between navigation, audio categories and playback. They allowed interface ideas to be considered before the group invested in detailed screens. This use of simple drawings is consistent with the value of low-fidelity prototypes for making design changes with limited preparation [5, Sec. "Benefits of Low-Fidelity Prototypes"].

V1 was a desktop-first, browser-based prototype implemented as a front end. Its main screens were Home, Search, Library, content detail and the full player. These screens supported connected task paths rather than only showing isolated layouts. Users could search the sample catalogue, open an item, change its saved state, find it in the Library and open unfinished spoken audio at its displayed progress. This level of interaction allowed the tests to examine labels, navigation choices, controls and feedback.

The catalogue and listening states represented the intended experience. The prototype did not provide a complete streaming service, so opening the player at an existing position was an interface response, not a test of audio delivery. V2 retained this implementation scope and the main task paths. It provided a revised interface for a second evaluation, while V1 remained available as the earlier version. Detailed screen descriptions and visual comparisons belong in the Design chapter.

3.6 Usability Testing

Two usability-testing rounds were conducted with three participants in each round. P1, P2 and P3 tested V1, while P4, P5 and P6 tested V2. All six were students, aged 20–26, who reported regular use of audio applications. Their backgrounds included mainly music listening and combinations of music, podcasts and audiobooks. These characteristics were relevant to the intended student audience. The records describe participants' listening habits but do not specify how they were recruited.

The evaluations used the interactive prototypes in a browser on a laptop, within the project's desktop-use focus. The V1 plan identifies Mittuniversitetet in Sundsvall as the location and specifies that the prototype ran locally. The V2 plan also specifies a laptop and browser. Participants' usual mobile or computer use was recorded as background information; the sessions did not constitute a comparison between devices.

The group divided moderation and observation between its members. Yacoub Dawli led the sessions and presented the tasks. Obid Alrhman Aljabi documented participant behaviour and comments, while Sarshar Chavoshi Rad recorded task times, clicks and navigation observations. This division allowed the moderator to attend to the participant while the observers recorded the interaction.

Each session began with an introduction to the university project and the purpose of the test. After information about participation and consent, the moderator asked which audio applications the participant normally used, which audio types they listened to, how

often they listened and whether they normally used mobile devices, computers or both. The three tasks were then presented individually. Participants were asked to explain what they were looking for, what they expected to happen and anything they found unclear while using the interface. This followed the think-aloud approach introduced in Section 2.5 [13, Sec. “Defining Thinking Aloud Testing”].

The instructions described the intended goal without specifying the control to select. For example, the unfinished-audiobook task did not tell participants to use Continue Listening. This left their choice of route available for observation. When prompting was needed, the moderator asked neutral questions about what the participant was thinking or expected. Participants were encouraged to try what made sense to them instead of being immediately directed to a solution.

Table 3.1 summarises the three core tasks and the completion criteria recorded in the test documentation. The first task included checking that the saved item could be found again. The second concerned returning to an existing listening position, while the third examined access to one content category. These tasks covered the main interaction priorities without requiring every prototype function to be tested.

Table 3.1: Core task goals and completion criteria used in both testing rounds.

Task	**Goal**	**Completion criterion**
1	Find and save Weezer for later.	The item is saved and can be found in the Library.
2	Find an unfinished audiobook and continue listening.	The player opens at the existing progress position.
3	Find where to browse podcasts.	The Podcasts category is reached.

The same core goals and completion criteria were used in the second round so that the group could compare the two versions without replacing the activities being examined. The wording allowed different valid routes to the goal. Finding content through a category, for example, could still meet the first task’s criterion. The records could therefore distinguish task completion from the particular route chosen.

The task records covered completion, elapsed time in seconds, click counts and navigation paths, alongside notes on hesitation, incorrect actions and any need for help. Relevant think-aloud comments accompanied these observations. Completion related to effectiveness, while time, clicks and routes provided information about effort and efficiency. Hesitation and comments helped examine whether controls were discoverable and whether their feedback was understood. These records supported interpretation of the measured values; a time or click count alone did not explain a difficulty.

Post-test questions covered the overall impression, easy or confusing parts, unclear labels, missing functions and suggested changes. The V2 records also included questions about Save and discoverability. The group reviewed these responses alongside the task observations in each round. This connected participants’ impressions to what happened during interaction. Individual accounts, measurements and the comparison of results are

presented in the Results chapter.

3.7 Iterative Evaluation

After the first evaluation, the group brought together task records, navigation paths and post-test comments to identify issues for redesign. The V1 analysis linked each identified issue to supporting observations, its effect on the tested task and a proposed action. The prioritised problem list distinguished concerns about reaching or discovering task-related content from smaller concerns about visual clarity. It also identified parts of the interface to retain.

The resulting redesign focused on Continue Listening visibility, the height of the Home introductory area and the clarity of the Save/Saved control. These areas connected the recorded concerns to specific changes that could be examined in another test. The basic navigation, categories, Library and core task goals were kept consistent. This limited the scope of the redesign and retained the main interaction context for comparison.

V2 was then evaluated with new participants using the procedure and tasks described in Section 3.6. Using different participants avoided repeating the tasks with people who had already learned the V1 paths. The second-round analysis considered the same completion criteria, measures, routes and participant feedback, including observations related to the targeted changes.

This comparison concerned the tested prototype interactions. With three different participants per round, differences between the records cannot be attributed solely to the redesign or generalised to all students. The purpose was to examine the revised design against the earlier concerns, without assuming that changing the interface established an improvement.

3.8 Research Ethics

The participant introductions explained that the group was evaluating the interface, not the person using it. They described the university project, the observation and written note-taking, and the intended use of the information. Participants were told that taking part was voluntary, that they could stop at any time and that they could raise questions or discomfort. The moderator then asked whether they were comfortable participating, and the records include agreement before the tasks. This approach addressed informed and voluntary participation as discussed in Section 2.6 and in Vetenskapsrådet's guidance [14, pp. 62–63].

The introductions stated that written notes would be anonymous and used for the university project. The retained records identify participants as P1–P6 instead of using personal names. Background information about listening habits and device use provided context for the tests without requiring names in the report.

Accurate reporting also informed the treatment of the material. The testing records

distinguish observations and participant comments from the group's interpretations. This distinction is maintained when describing the method and reporting the results, consistent with the responsibility to communicate research honestly [14, pp. 11–12].

Chapter 4

Design

4.1 Initial Design Alternatives

The early concept considered separate audio categories and a mixed feed with filters. Categories give listeners a clear starting point when they already know which audio type they want. A mixed feed supports browsing different types together, but needs clear labels and filters to distinguish them.

Cards and lists offered another choice. Cards leave room for artwork and visual recognition, but fewer items fit on the screen. Lists show titles, creators and other information more compactly. Their suitability depends on whether the user is browsing or looking for a particular item.

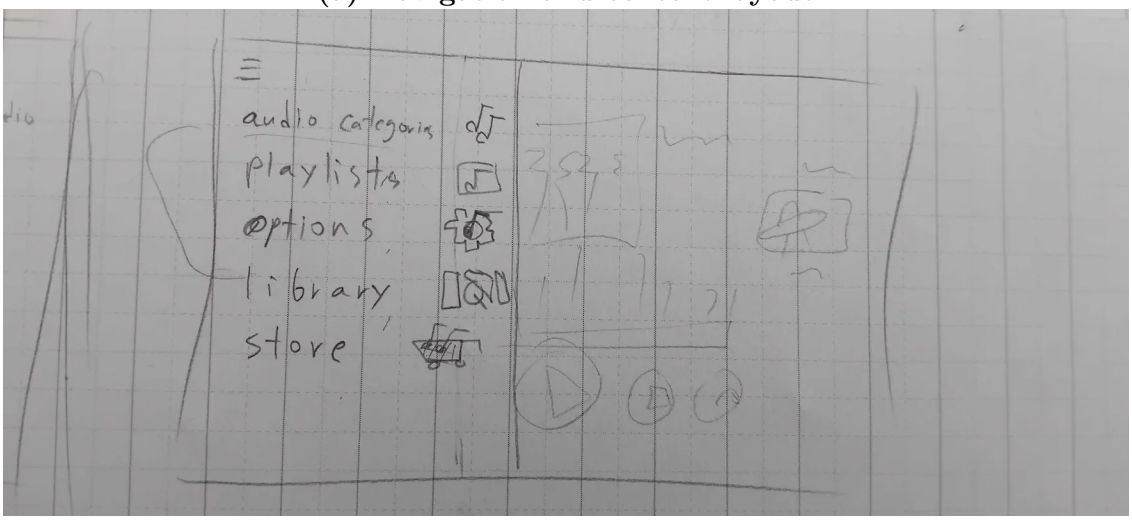
The chosen interface combines these ideas: cards in separate Home sections, a list of search results and one Library with content filters. Each audio category has a browsing route, while saving follows a shared pattern. This gave us a starting design that we could test and improve.

4.2 Lo-Fi Sketches

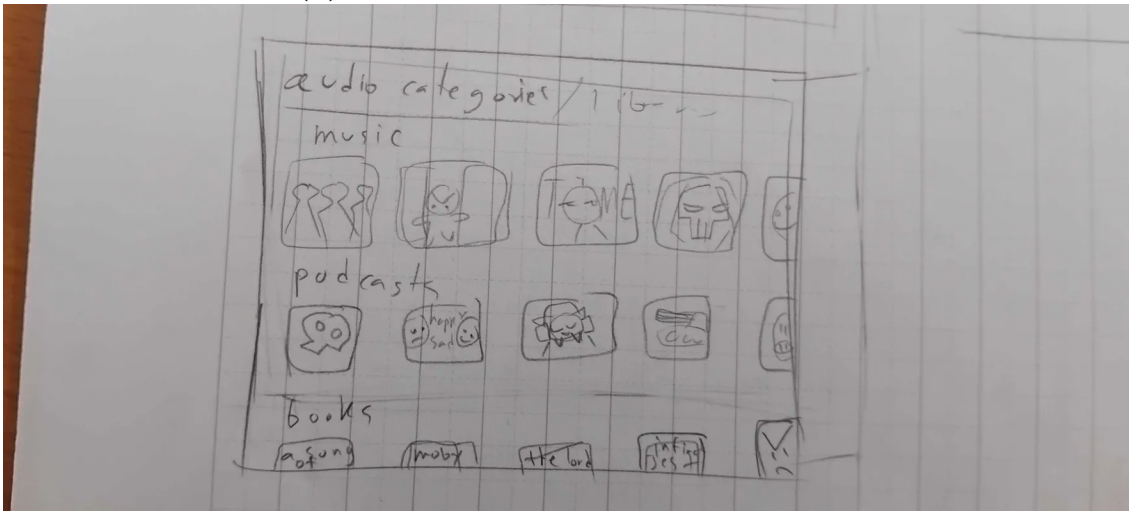
Figure 4.1 presents the group's three original pencil sketches. Sketch (a) places a labelled navigation area on the left, with entries for audio categories, playlists, options, Library and a store. The area on the right contains rough content blocks and playback symbols. Sketch (b) concentrates on browsing: rows labelled music, podcasts and books contain small drawings of covers. Sketch (c) explores a player around Weezer, with central artwork, a large play symbol, neighbouring controls and a horizontal volume control below.

These drawings made the layout ideas visible before the interactive screens were built. They helped the group consider how much space navigation, content and playback would need. Their rough appearance also left room to change the arrangement without first producing detailed visual designs. They were starting points rather than specifications: for example, the store shown in sketch (a) was not included in the tested prototype.

(a) Navigation and content layout



(b) Audio categories and cover cards



(c) Player layout

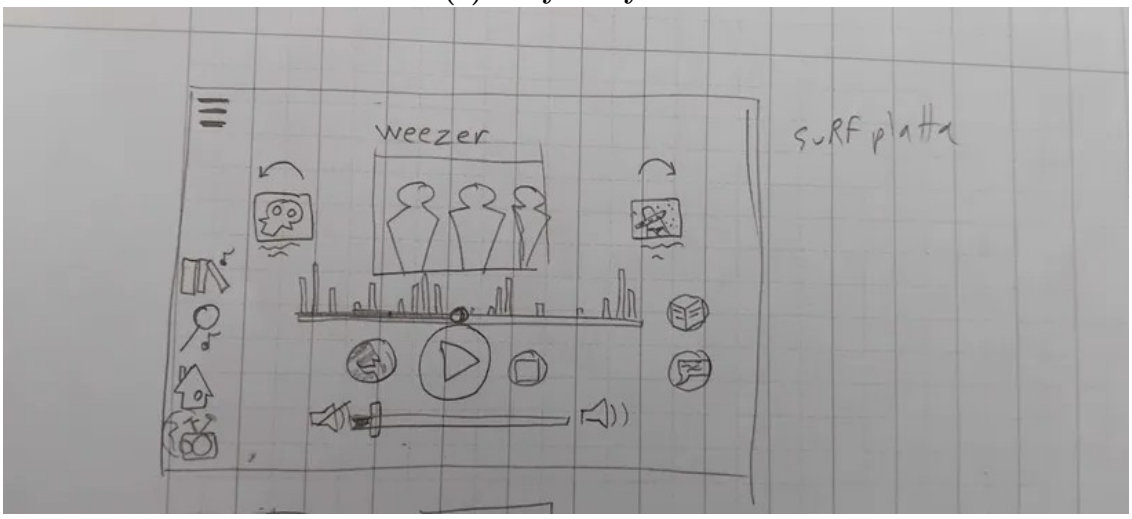


Figure 4.1: The three original project sketches, exploring navigation, category rows and playback. The photographs retain their original content and proportions.

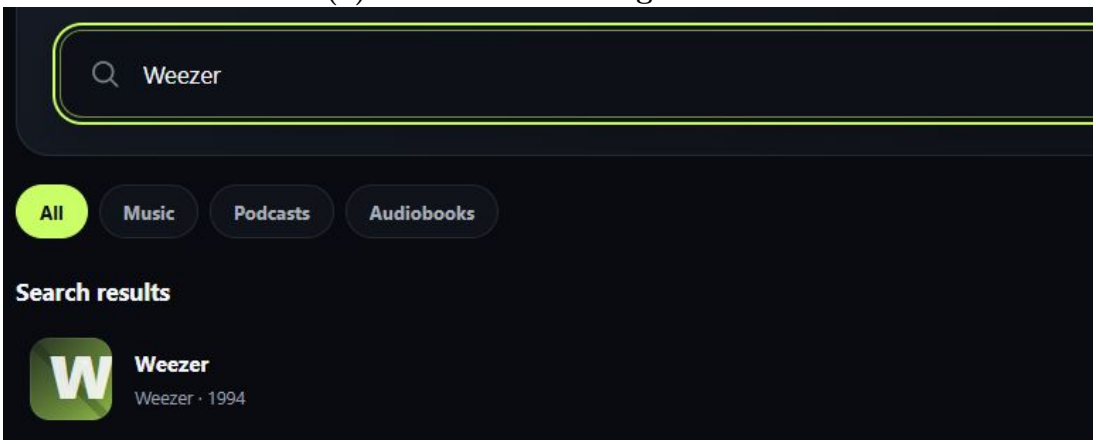
4.3 Omnisound Interface Structure

Omnisound has five main screen types: Home, Search, Library, content detail and the full player. Persistent navigation provides Home, Search and Library, followed by separate Music, Podcasts and Audiobooks routes. These remain available when opening content or playback.

Home includes Continue Listening for unfinished spoken audio. Search covers the sample catalogue, with type filters. Library collects saved items and offers similar filters. Details provide Save and Play or Resume. The full player groups listening controls; a smaller player remains available during browsing after playback opens.

Figure 4.2 shows Weezer in Search and, after saving, in Library. Results use lists for scanning, while Library uses covers and titles for recognition. Shared destinations and controls connect the audio types without requiring different saving and playback patterns.

(a) V1 Search: finding Weezer



(b) V1 Library: Weezer after saving

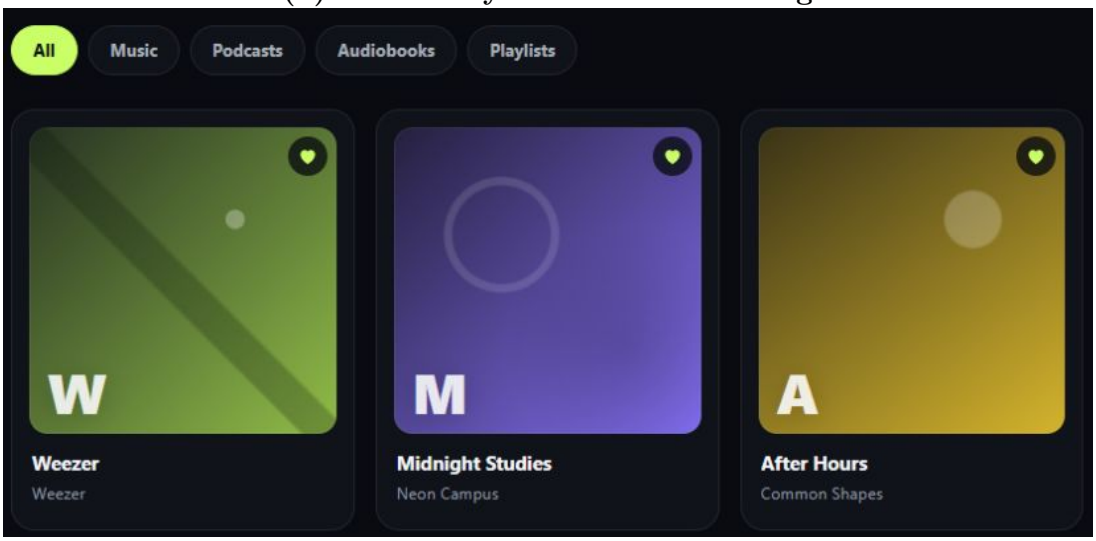


Figure 4.2: V1 Search and Library, cropped to show the query, filters, result and saved collection.

4.4 Hi-Fi Prototype V1

V1 developed the sketches into the first interactive high-fidelity prototype. Instead of viewing isolated drawings, users could follow the main activities from one screen to another. The category rows and artwork remained part of the design, while labelled buttons and visible states made searching, saving and resuming available for testing.

The Home screen in Figure 4.3 starts with a large introductory banner. Below it, a small Continue Listening label sits above the heading "Back in progress". The unfinished items show their audio type, title, creator and listening progress. Separate Music, Podcasts and Audiobooks sections follow further down the page. The sidebar provides a direct route to each category even when its Home section is below the visible area.

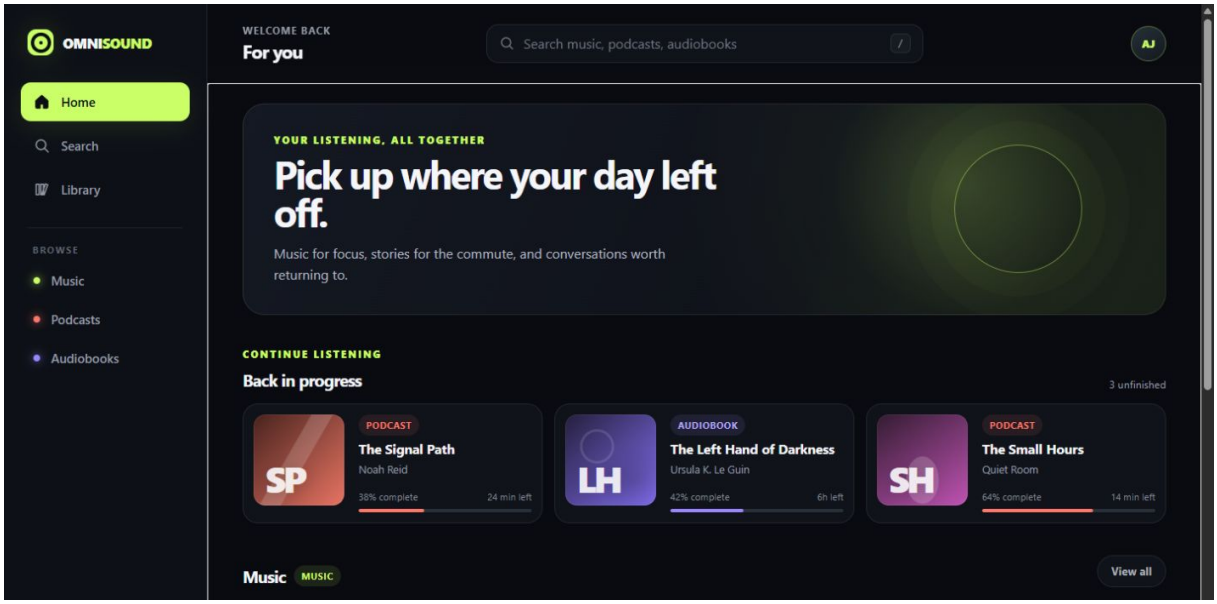
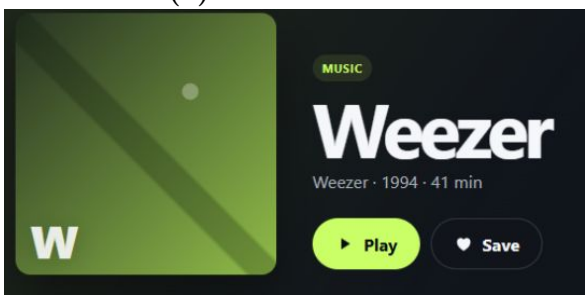


Figure 4.3: The V1 Home screen, showing the persistent navigation, large introductory banner and unfinished content. The capture shows the upper part of Home; further category rows continue below.

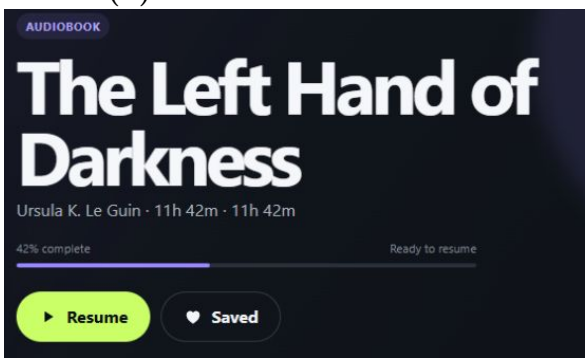
One important workflow begins by searching for Weezer and opening the result. Its detail screen places Save beside Play, as shown in Figure 4.4(a). Selecting Save changes the label to Saved and briefly displays "Weezer saved to your Library". Opening Library then shows the saved item, as in Figure 4.2(b). The connection matters because saving should provide a route back to the content, rather than only changing a button on the current screen.

The other main workflow returns to unfinished audio. A user can open *The Left Hand of Darkness* from Continue Listening and recognise its existing progress on the detail screen. Resume then opens the full player at that displayed position. Figure 4.4(b–c) shows these connected states. The player includes progress, play/pause, volume and spoken-audio speed controls. Its sample content and progress represent interface states; they do not demonstrate a complete streaming service.

(a) Weezer detail



(b) Unfinished audiobook



(c) Full player after Resume



Figure 4.4: V1 saving and resuming workflows: Save beside Play on the Weezer detail page, progress and Resume on the audiobook detail page, and the resulting full-player view. Crops focus on the relevant content and controls.

4.5 Design Decisions and HCI Connections

The principles in Section 2.3 guided concrete parts of the interface. The sidebar labels and categories in Figure 4.3 support visibility. Labels such as Music and Audiobook explain what an item is before it is opened, without relying only on artwork or colour.

Feedback connects actions to outcomes. Saving changes the label and displays a confirmation naming Library. Navigation highlights show the current destination, while play/pause controls reflect the player state. However, clear feedback after saving does not ensure that Save is easy to find beforehand, as the first test showed.

Consistency links the audio types. Their cards open similarly arranged detail pages, with Save beside Play or Resume. The lists below contain tracks, episodes or chapters as appropriate, while the main interaction pattern stays familiar. This connects expectations and behaviour as discussed in Section 2.4.

Recognition also shaped the screens. Titles and covers help users identify Library items, and progress bars identify unfinished audio. These cues reduce the need to remember exact titles or listening positions. Buttons such as Play, Resume and Save use familiar icons and clear labels to show users what they do. Play and Resume state what will happen; the heart with Save gives the icon a written meaning. The redesign addressed cues that needed more emphasis.

4.6 Redesign Decisions

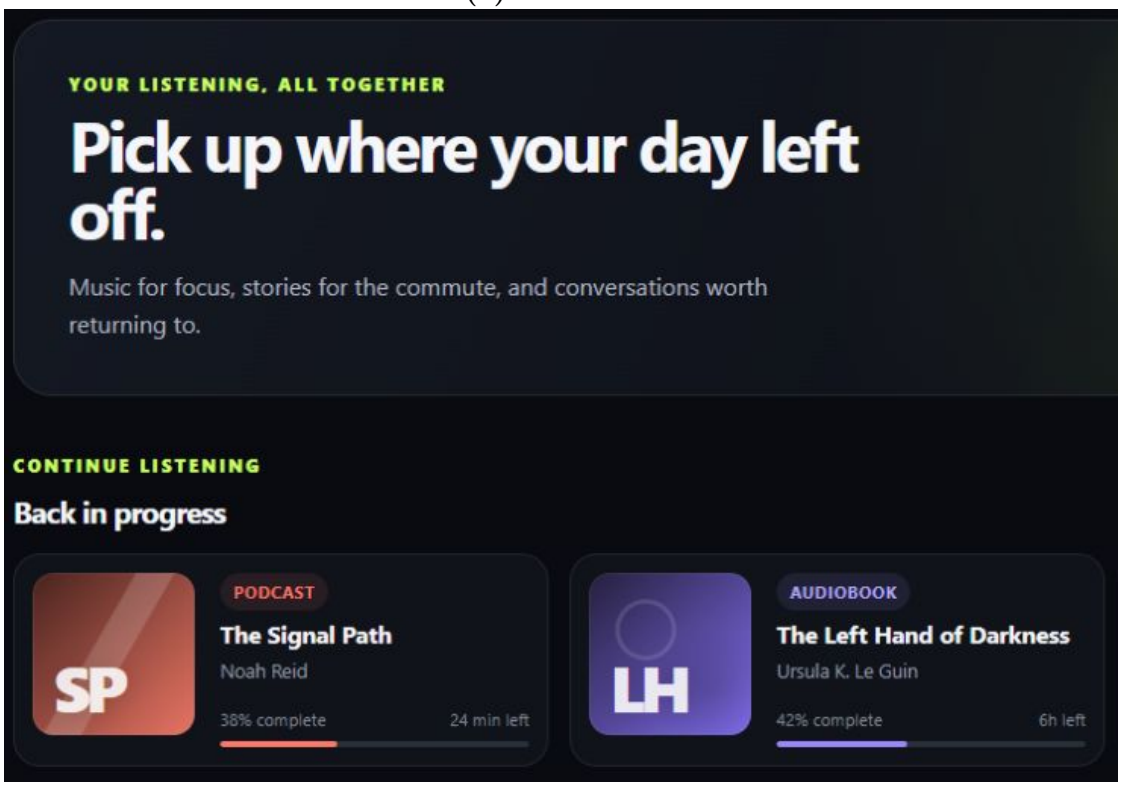
The first usability test showed that some participants looked for unfinished audio through Audiobooks or Library instead of Continue Listening. Those routes worked, but the intended Home shortcut was being overlooked. The group therefore focused the redesign on making that section easier to notice, reducing the space taken by the introduction and clarifying the saving control.

Figure 4.5 shows the Home changes. V2 reduces the banner's height and the space before unfinished content. It also replaces the small Continue Listening label and "Back in progress" heading with a single, larger Continue Listening heading and a lime marker. The unfinished cards and their progress information remain. The intention was to bring attention to content that users could return to, while keeping a short introduction to Omnisound.

The Save control received a smaller change. The first test recorded a brief hesitation in locating it beside Play, so V2 strengthened its border and gave the unsaved and saved states more distinct appearances. The label and confirmation message were retained because they explained the action's result. Figure 4.6 shows the revised states.

The group kept the sidebar, Search, Library and audio categories because the first evaluation supported these routes. Library also provided another way to reach unfinished content. We then tested V2 to see whether these changes made the interface easier to use.

(a) V1 Home



(b) V2 Home

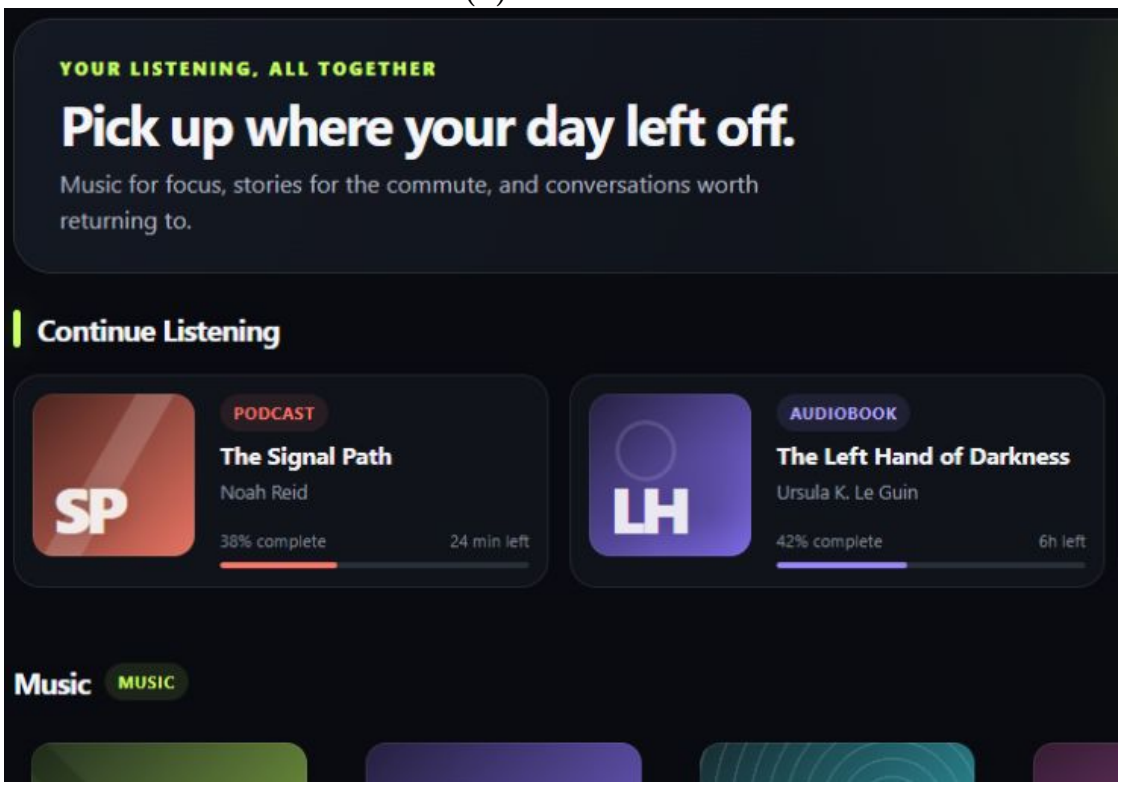


Figure 4.5: Home before and after the redesign. Both prototypes were captured at a 1440×900 viewport and 100% zoom, then cropped to the same region of the banner and first two unfinished cards, at the same scale. The smaller banner and revised heading bring unfinished content higher on V2 Home.

4.7 Hi-Fi Prototype V2

V2 keeps the five main screen types and navigation structure of V1. The Home comparison in Figure 4.5 also shows the final arrangement: the introduction occupies less space, and the more direct Continue Listening heading gives unfinished content greater emphasis. Users can still search, browse categories or open Library through the same destinations.

On the detail page, the unsaved control uses an outlined heart and the label Save. After saving, it shows a filled heart, lime styling and Saved. The comparison in Figure 4.6 captures the same Weezer page before and after selecting Save. The control stays beside Play, so its position remains familiar while its changed appearance communicates the current state. The brief Library confirmation is also retained, giving immediate feedback as well as a state that remains visible on the page.

(a) Before saving **(b) After saving**

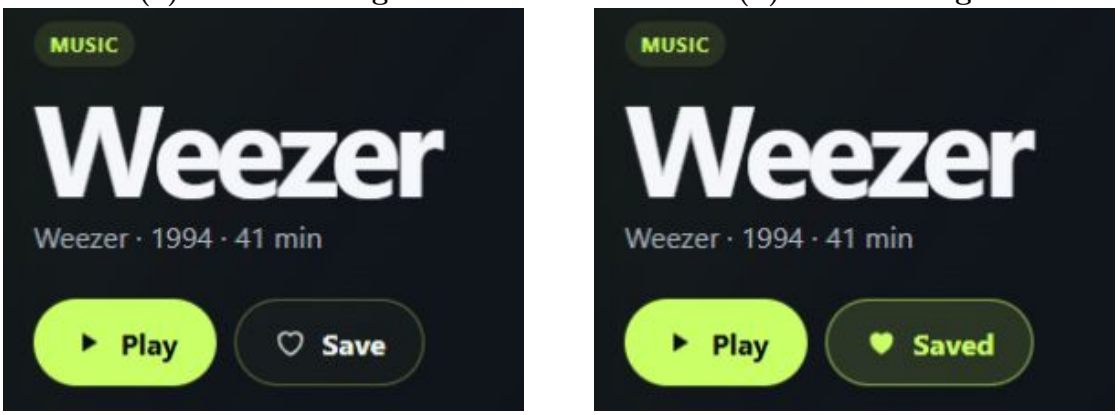


Figure 4.6: The Save and Saved states in V2, captured before and after saving the same item. Matching crops preserve the detail context; only the saving control changes. The temporary confirmation has disappeared in the second capture.

The group evaluated V2 in a second usability test using the same core tasks: finding and saving content, resuming unfinished audio and browsing podcasts. This completed the sequence from V1 evaluation to redesign and another test. Chapter 5 presents the recorded outcomes and comparison between the two rounds.

Additional screenshots from both prototypes are provided in Appendix A.

Chapter 5

Results

5.1 Data Collection Results

Six students took part in the usability tests: P1–P3 used V1 and P4–P6 used V2. All six reported using audio applications every day, and Spotify appeared in every participant's account. Other applications included YouTube, YouTube Music, Audible, Pocket Casts, Apple Podcasts and Storytel. Table 5.1 summarises their reported preferences, applications and usual devices.

Music was part of all six participants' listening habits. P1 and P4 also listened to podcasts, while P3 described podcast listening as rare. P2, P5 and P6 mainly listened to music and audiobooks. The group therefore included both mainly music listeners and people who regularly combined audio types.

Table 5.1: Reported audio preferences, applications and usual devices.

ID	**Round**	**Audio preferences**	**Usual applications**	**Usual devices**
P1	V1	Mainly music; some podcasts	Spotify, YouTube	iPhone, laptop
P2	V1	Music and audiobooks	Spotify, Audible, Pocket Casts	Android phone, Windows PC
P3	V1	Mainly music; rare podcasts	YouTube Music, Spotify	Android phone
P4	V2	Mainly music; some podcasts	Spotify, YouTube Music	Android phone, Windows laptop
P5	V2	Music and audiobooks	Spotify, Apple Podcasts, Storytel	iPhone, MacBook
P6	V2	Music and audiobooks	Spotify, Audible	Android phone, Windows laptop

Phone use was common, but several participants also listened on computers. P1 and P4 mentioned computer use while studying. P2 used both platforms for music and mainly a phone for audiobooks. P3, P5 and P6 mainly used phones.

5.2 Identified User Needs

The project concept already prioritised finding content, saving it in a shared Library and returning to unfinished listening. Keeping music, podcasts and audiobooks recognisable was another starting requirement. The testing records provided observations about how participants used these functions and what they found unclear.

Finding and retrieving content involved more than one route. Participants used Search or Music to locate Weezer, then saved the item and checked Library. Their actions showed how the original finding-and-saving goal was used in practice. Library also served another purpose for P3, who looked there for the unfinished audiobook. Saved collections and unfinished listening therefore overlapped in this participant's route through V1.

The feedback also identified specific priorities within the concept. P2 and P3 did not use Continue Listening in the first test and later said they had not noticed it. P1 suggested making Save more obvious. In V2, participants recognised the changed Saved state and checked the item in Library. These observations concerned finding the relevant control and understanding what happened after selecting it. Category names were also used directly during the tasks, and participants' post-test comments described the separation between audio types as understandable.

5.3 Usability Test 1 Results

P1, P2 and P3 completed all three V1 tasks: finding and saving Weezer, resuming an unfinished audiobook and browsing podcasts. This gave nine completed tasks out of nine attempts, with no direct assistance from the test leader. Table 5.2 presents the recorded completion times and outcomes.

Table 5.2: V1 task times in seconds and completion results.

Task	**P1**	**P2**	**P3**	**Completed**	**Direct help**
Find and save Weezer	31	24	28	3/3	None
Resume audiobook	19	27	35	3/3	None
Browse podcasts	4	5	7	3/3	None

For the first task, P1 selected Search in the sidebar, while P2 opened Search through the top search field. P3 opened Music and found Weezer among the visible cards. Each participant opened its details, selected Save and found the item in Library. Their routes differed, but all reached the requested saved collection.

P1 noticed Play before Save and briefly paused before locating the saving control. After saving, P1 understood the confirmation and opened Library to check the result. P3 initially assumed the heart would add the item somewhere, then named Library after reading the confirmation message. In the post-test response, P1 suggested: "Maybe make Save slightly more obvious." No participant needed help to complete the saving task.

The audiobook task produced the largest spread in V1 times, from 19 to 35 seconds. P1 used Home, Continue Listening, the audiobook details and Resume. P2 instead opened Audiobooks from Home, selected *The Left Hand of Darkness* and chose Resume. P3 went from Home to Library, opened the same audiobook and selected Resume. The player opened at the existing progress in all three cases.

Only P1 used Continue Listening directly. P2 scanned the audiobook cards through the category route, while P3 passed the Continue Listening section and searched in Library. P3 said during the task: "Maybe unfinished things are saved here." The progress information and Resume control then identified the audiobook as unfinished. The recorded resume-task click counts were two for P1, three for P2 and four for P3.

All three participants reached Podcasts. P1 and P2 selected it directly in the sidebar. P3 accidentally selected Music first, noticed its active highlight and immediately switched to Podcasts. This was the recorded wrong action in that task; P3 still completed it without help.

Post-test comments matched the different audiobook routes. P2 said: "I didn't immediately notice Continue Listening. I went to Audiobooks instead." P3 also said the section had not been noticed and commented: "Maybe make Continue Listening more obvious. The big banner above it takes a lot of space." The main V1 observations were completed tasks through several routes, an overlooked Home shortcut for P2 and P3, a brief hesitation around Save for P1 and one category-selection error followed by immediate recovery.

5.4 Usability Test 2 Results

P4, P5 and P6 completed the same three core tasks using V2. All nine task attempts were completed without direct assistance. Their recorded times are shown in Table 5.3, using the same layout as the V1 results.

Table 5.3: V2 task times in seconds and completion results.

Task	**P4**	**P5**	**P6**	**Completed**	**Direct help**
Find and save Weezer	22	24	21	3/3	None
Resume audiobook	13	15	12	3/3	None
Browse podcasts	4	4	3	3/3	None

P4 used the top search field to find Weezer, and P5 selected Search in the sidebar. P6 opened Music and found the item directly. All three selected Save on the detail page,

recognised the Saved state and opened Library to check that Weezer appeared there. The music-category route therefore remained available alongside Search in the second round.

P4 noticed Save immediately and described it as obvious. P5 briefly looked between Play and Save before selecting it; the notes record a pause of approximately one second. After the button changed to Saved, P5 said: "Okay, now I know it worked." P6 selected Save without recorded hesitation.

All three participants used Continue Listening to resume the unfinished audiobook. Their route was Home, Continue Listening, the audiobook details and Resume. Each then reached the player at the existing position. P4 looked below the banner and selected the audiobook, P5 saw the section without scrolling, and P6 referred to the displayed progress when identifying unfinished content. The notes record two clicks and no hesitation for each participant on this task.

For podcast browsing, P4, P5 and P6 each selected Podcasts in the sidebar. The recorded times were four, four and three seconds respectively, with one click each. No wrong category selections or difficulties were recorded for this task in V2.

The post-test responses referred to both the targeted controls and the general navigation. P4 described the sidebar and searching as the easiest parts and said: "No, Continue Listening was pretty visible." P4 also said the heart and the word Save made the action clear. P5 named Continue Listening and navigation as the easiest parts, while P6 named categories and Continue Listening. All three described Save as understandable, and none reported a confusing label.

The feedback also included suggestions beyond the tested tasks. P5 wanted Recently Played as a possible later feature, and P6 suggested sorting the Library. Neither suggestion described an obstacle that prevented task completion. The V2 records therefore contain successful task outcomes, direct audiobook-resumption routes and a brief saving pause, alongside ideas for additional functions.

5.5 Comparison Between Iterations

Task completion was the same in both rounds: nine out of nine attempts, with no direct help required. The recorded differences concerned time, route choice and hesitation rather than additional completed tasks. V1 participants took between 24 and 31 seconds to find and save Weezer, compared with 21–24 seconds in V2. Audiobook resumption took 19–35 seconds in V1 and 12–15 seconds in V2. Podcast browsing took 4–7 seconds in V1 and 3–4 seconds in V2.

Figure 5.1 compares the average completion times. Each average uses the three participant times for that task in its round. The means are 27.7 seconds for finding and saving in V1 and 22.3 seconds in V2; 27.0 and 13.3 seconds for resuming the audiobook; and 5.3 and 3.7 seconds for browsing podcasts. Values are rounded to one decimal place.

The largest difference in average time was in audiobook resumption. Route choice also differed on that task: one of the three V1 participants used Continue Listening directly,

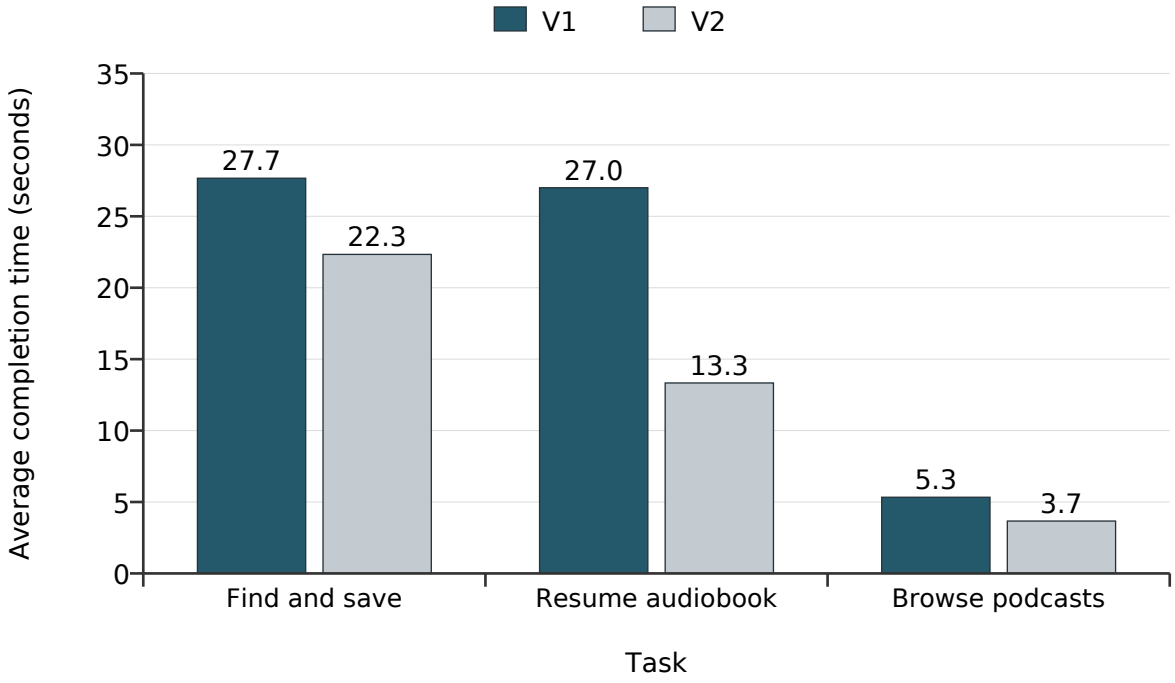


Figure 5.1: Average task completion times, based on three participants per testing round, with different participants in V1 and V2. Bar labels show means rounded to one decimal place.

compared with all three V2 participants. The other V1 participants still completed the task through Audiobooks or Library. Finding Weezer through Music occurred in both rounds, alongside the two entry points to Search.

The saving records include a hesitation for P1 in V1 and a shorter pause for P5 in V2. Both completed the task and checked Library. Category navigation also reached the requested destination in both rounds; P3's initial Music selection in V1 was followed by immediate recovery, while no such wrong selection was recorded in V2.

The recorded completion times were lower in the second round, especially for resuming the audiobook, and Continue Listening was used more often. These are comparisons between two groups of three different participants; the lower times alone do not show that the redesign caused the difference. Across both rounds, all participants completed the tested tasks without direct assistance.

Chapter 6

Discussion

6.1 Interpretation of Results

Both versions supported the main tasks for the participants who tested them. Tables 5.2 and 5.3 show nine out of nine completed tasks in each round, without direct assistance. This suggests that the tested navigation and core interactions were understandable. However, reaching a goal did not mean that every route was equally easy to find.

The clearest difference concerned the unfinished audiobook. Only one of the three V1 participants used Continue Listening directly; the others went through Audiobooks or Library. All three V2 participants used Continue Listening, and the average time fell from 27.0 to 13.3 seconds (Figure 5.1). The second group found unfinished content more directly and completed the task faster. The updated Home screen may therefore have made the intended shortcut easier to notice.

Saving showed a smaller change. P1 briefly hesitated before finding Save in V1. All V2 participants understood the Save/Saved state, although P5 briefly paused before saving. The clearer button appearance and retained confirmation may have helped communicate both the action and its result. The states shown in Figure 4.6 provide a visible reminder after the temporary confirmation disappears.

V1 already offered useful ways to complete the tasks. Search, Library and the categories supported several successful routes, including finding Weezer through Music. P3 also recovered immediately after selecting the wrong category. Keeping these routes while changing a few specific areas was a reasonable response to the findings. The remaining brief pause in V2 is a reason to keep observing the saving control, rather than assume that every small difficulty has disappeared.

6.2 Connection Between Theory and Findings

The Home findings connect visibility with attention, as discussed in Sections 2.3 and 2.4. Continue Listening was available in V1, but its presence alone did not make it an obvious starting point for P2 and P3. The large banner occupied space above it. Reducing the

banner and using a more direct heading gave unfinished content greater prominence, as Figure 4.5 shows. This offers a plausible explanation for the different routes in V2, without treating appearance as a complete explanation of behaviour.

Feedback mattered after an action as well. The changed Saved label and the message naming Library told participants what had happened and where the item could be found. P3's move to Library after reading the confirmation is a concrete example. The active navigation highlight also helped P3 recognise an accidental category selection. These responses supported the next decision instead of leaving the participant to guess the current state.

Consistency and recognition help explain why the alternative routes still worked. Category names, titles and progress information gave participants something to recognise. P3 expected unfinished content to be in Library and could identify the audiobook there. Shared controls across the audio types supported that route. Keeping the navigation and Library structure in V2 preserved these interaction patterns alongside the clearer Home shortcut.

The iteration also put the user-centred approach in Section 2.1 into practice. We chose the V2 changes from observed hesitation, route choices and comments. This gave each change a specific problem to address and a reason for another test. Personal preferences about the appearance could not have told us whether listeners would notice Continue Listening during a task.

6.3 Strengths and Limitations

A strength of the project was the connection between its stages. Early sketches led to interactive prototypes, and the first test led to focused changes that were tested again. The same core tasks gave both rounds a common basis for comparison. Retaining V1 also made the design changes visible. Together, the observations, comments, times and routes provide a clearer account of the interaction than a finished interface alone would.

Only three participants tested each version, so the findings describe a small set of experiences. Different people took part in the two rounds. This avoided repeating the tasks with participants who had already learned the V1 routes, but differences between people could also affect times and choices. We therefore cannot attribute the lower V2 times only to the redesign. The Home changes were introduced together, so the tests also do not separate the effect of the smaller banner from the clearer heading.

The tasks covered finding and saving content, resuming an audiobook and browsing podcasts. They did not examine every function or repeated use over time. Testing took place in a laptop/browser setting, although many participants normally listened on phones. The results therefore give little information about the full mobile experience. The sample catalogue and displayed progress also limited what could be tested: real audio delivery, large collections, subscriptions and account handling were outside the front-end prototype.

The initial interview plan is retained, but completed initial interview responses are not. This limits how closely the early requirements can be traced to interview findings. The concept and later testing still identify useful priorities for the prototype, but they provide a narrower view than a fuller record of students' needs would. Our first goal was therefore addressed within the selected listening activities, rather than through a complete account of everything students might need from an audio service.

6.4 Evaluation of Methods

Starting with sketches was useful for considering navigation, content and playback before building detailed screens. It left room to change the arrangement without first completing an interactive version. The later prototypes served a different purpose: participants could actually search, save, browse and resume content. We could observe the connection between screens and the result of an action, rather than rely only on opinions about static layouts.

Think-aloud comments helped explain choices that the completion totals would have hidden. P2 looked in Audiobooks and P3 expected unfinished content in Library. Their comments and routes showed why the intended Home shortcut needed attention. Task times and click counts added information about the effort involved, while the observations showed where participants paused or changed direction. Combining these records was more useful than treating completion or time as the whole result.

The division between moderation and observation also helped the group follow the interaction while recording it. Explaining voluntary participation and using participant labels also addressed consent and privacy, as described in Section 3.8. In later work, we would retain these practices and make the link between interview notes, requirements and design decisions easier to follow. A larger and more varied participant group, with consistent task wording and test conditions, would also give us a broader basis for reviewing the interface.

6.5 Future Work

The next evaluation should include mobile interaction and a wider range of listening habits. It should also ask people to return to saved or unfinished audio after a break or over several visits. This would examine whether Library and Continue Listening remain useful when users have to recognise content later, rather than immediately after a task. More realistic playback and retained progress across visits would support that evaluation.

P5's suggestion of Recently Played and P6's suggestion of Library sorting are reasonable ideas to explore. They were not needed to complete the tested tasks, so they should be considered alongside further observations of how people return to content. A small prototype extension could examine whether these options help with that activity and whether their purpose is clear alongside saved and unfinished items.

6.6 Conclusion

Omnisound explored how music, podcasts and audiobooks could share search, saving and playback while keeping their content types recognisable. The project identified a focused set of interaction priorities, explored alternative layouts through early ideas and sketches, and developed two interactive prototypes. This met the design and evaluation goals and helped us understand how people used the main functions.

All participants completed the three core tasks in both usability-testing rounds. The first evaluation nevertheless showed that Continue Listening could be easier to notice and Save could be more prominent. We made focused changes to Home and saving in V2, then tested the same goals again. The second group used Continue Listening more directly and recorded shorter task times. These findings support the value of the iteration for the tested interactions, while remaining specific to the small groups and prototype scope.

The main lesson was that making a function available does not guarantee that people will find it when they need it. Watching participants use Omnisound revealed difficulties that were easy to overlook while designing the screens. Their actions and comments gave us concrete reasons to make changes and a clear focus for testing the next version.

References

[1] Spotify, "Spotify expands audiobook access for premium subscribers in 5 more European countries," Spotify Newsroom, Nov. 2025, published November 18, 2025. Accessed October 13, 2026. [Online]. Available: https://newsroom.spotify.com/2025-11-18/audiobooks-in-premium-sweden-denmark-finland-iceland-monaco/

[2] Apple, "Apps by Apple," Apple Official Website, accessed October 13, 2026. [Online]. Available: https://www.apple.com/apps/

[3] J. Åhlander, "Projekt – användargränssnitt: Människa-datorinteraktion," Mid Sweden University, course project instructions, Oct. 2023, dated October 10, 2023; local file: P3 Projekt.pdf.

[4] Y. Rogers, H. Sharp, and J. Preece, *Interaction Design: Beyond Human-Computer Interaction*, 6th ed. Wiley, 2023, chapter 1 publisher excerpt. Accessed October 13, 2026. [Online]. Available: https://media.wiley.com/product_data/excerpt/9X/11199010/111990109X-20.pdf

[5] K. Pernice, "UX prototypes: Low fidelity vs. high fidelity," Nielsen Norman Group, Dec. 2016, published December 18, 2016. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/ux-prototype-hi-lo-fidelity/

[6] International Organization for Standardization, "Systems and software engineering—systems and software quality requirements and evaluation (SQuaRE)—general framework for common industry format (CIF) for usability-related information," International Organization for Standardization, Technical Report ISO/TR 25060:2023, 2023, public preview, Section 3.1. Accessed October 13, 2026. [Online]. Available: https://www.iso.org/obp/ui?_escaped_fragment_=iso:std:iso:tr:25060:ed-1:v1:en

[7] J. Nielsen, "10 usability heuristics for user interface design," Nielsen Norman Group, Apr. 1994, published April 24, 1994; last reviewed January 30, 2024. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/ten-usability-heuristics/

[8] D. Norman, "Signifiers, not affordances," JND.org, author's version of an article in ACM Interactions, vol. 15, no. 6, Nov. 2008, posted November 17, 2008. Accessed October 13, 2026. [Online]. Available: https://jnd.org/signifiers-not-affordances/

[9] M. Chan, “Mental models,” Nielsen Norman Group, Jan. 2024, published January 26, 2024; last reviewed September 1, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/mental-models/

[10] M. Rosala and K. Pernice, “User interviews 101,” Nielsen Norman Group, Sep. 2023, published September 17, 2023; last reviewed July 15, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/user-interviews/

[11] R. Krause and K. Pernice, “Affinity diagramming for collaboratively sorting UX findings and design ideas,” Nielsen Norman Group, Apr. 2024, published April 26, 2024; last reviewed August 7, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/affinity-diagram/

[12] P. Laubheimer, “3 persona types: Lightweight, qualitative, and statistical,” Nielsen Norman Group, Jun. 2020, published June 21, 2020. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/persona-types/

[13] J. Nielsen, “Thinking aloud: The #1 usability tool,” Nielsen Norman Group, Jan. 2012, published January 15, 2012. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/thinking-aloud-the-1-usability-tool/

[14] Vetenskapsrådet, “God forskningssed 2024,” Vetenskapsrådet, Tech. Rep. VR2405, 2024, accessed October 13, 2026. [Online]. Available: https://www.vr.se/analys/rapporter/vara-rapporter/2024-10-02-god-forskningssed-2024.html

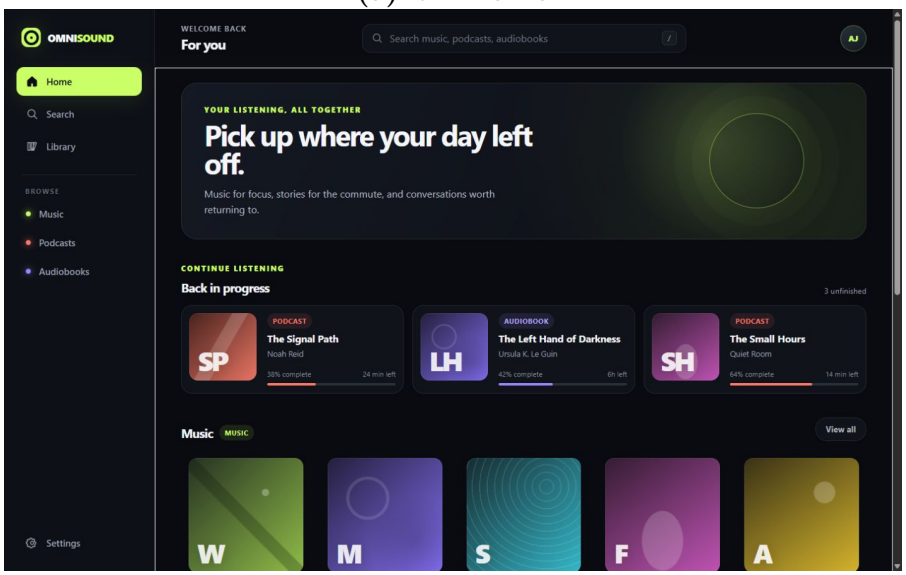
Appendix A: Additional Prototype Screens

This appendix shows additional screenshots from the two Omnisound prototypes. Chapter 4 focuses on particular controls and design changes, while these images provide a wider view of the interfaces.

A.1 Home Screens

Figure A.1 shows the Home screens from V1 and V2. Both versions include navigation and audio categories, while V2 places more emphasis on Continue Listening.

(a) V1 Home



(b) V2 Home

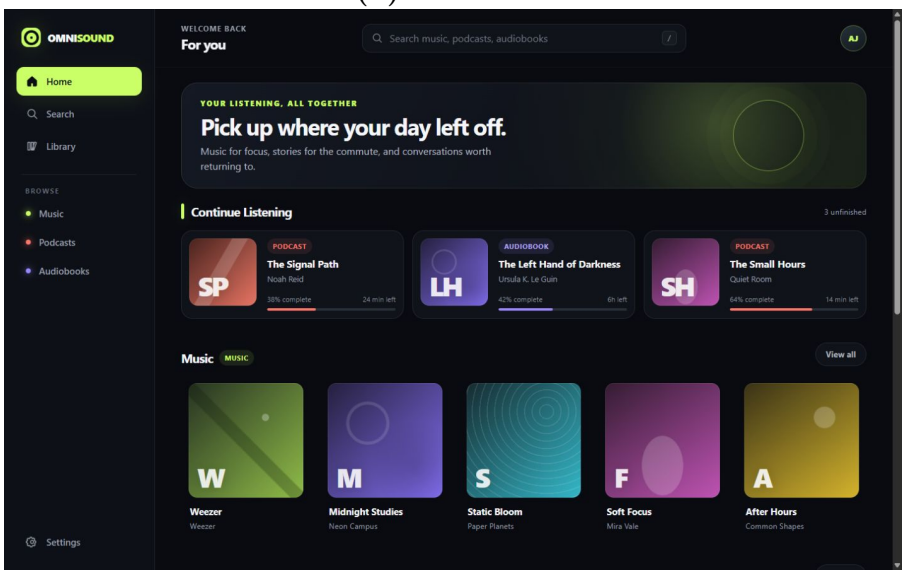
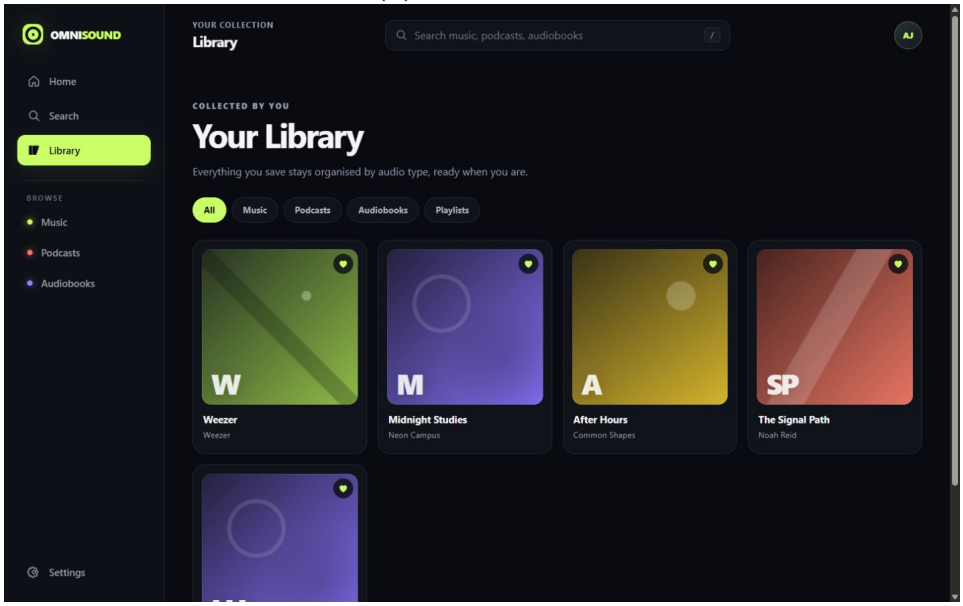


Figure A.1: Home screens from the V1 and V2 prototypes.

A.2 Library and Saving

Figure A.2 shows the V1 Library screen and a V2 content detail screen after an item has been saved. These are wider views of the interfaces discussed in the main report.

(a) V1 Library



(b) V2 Saved item detail

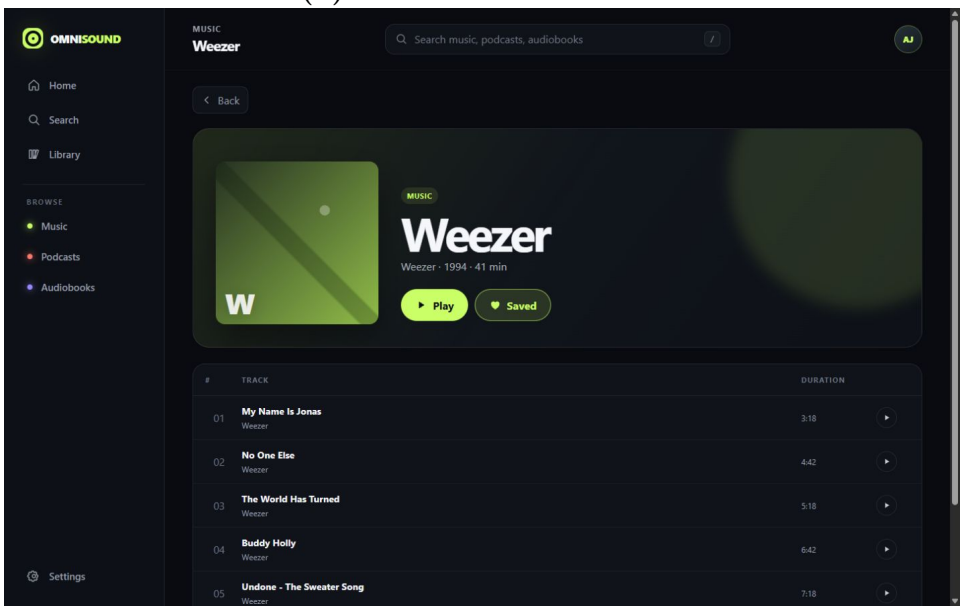


Figure A.2: Additional interface views from V1 and V2.

Appendix B

Usability Testing Documentation

This appendix includes the complete V1 and V2 usability-testing documents so that the test procedures, participant observations and measurements can be read together with the report.

The group has confirmed that both testing rounds took place in person and that the observations, times and comments are its completed testing records. The source pages still contain template wording such as “simulated” and “fictional example”, along with drafting instructions. These have been retained rather than editing the original documents.

Some interpretations in the notes use stronger wording about improvement than the main report. Chapters 5 and 6 compare two small groups of different participants and do not establish that the redesign alone caused the differences.

Participants are identified as P1–P6. The original pages retain ages, listening habits and device information. The V2 document also contains a local prototype file reference.

- V1 usability testing: the complete 19-page document for the first testing round, with P1–P3.
- V2 usability testing: the complete 18-page document for the second testing round, with P4–P6.

The original document page numbers are retained. The report’s page numbers continue at the bottom centre of each page.

OMNISOUND · HCI PROJECT · GROUP 1

V1 Usability Testing

Step 6 · Initial usability evaluation · English-language Notion notes

> **Source note:** This PDF is a formatted conversion of the supplied Notion text. It preserves its statements identifying participant accounts and measurements as **simulated examples**. The PDF is not independent verification of participant testing.

Absolutely. Below is a **complete simulated example of Step 6** for Omnisound, filled out from beginning to end so you can see exactly what a real usability-testing round could look like.

Important: all participants, answers, timings, observations and findings below are fictional examples. Use this to understand the process and structure. For the actual project, replace them with what your real test participants actually do and say.

The structure follows Session 5: plan/recruit/prepare and consent → pre-test questions → usability tasks using think-aloud → post-test questions → analysis → redesign. ProjectSession5_20251009_Usabil… ProjectSession5_20251009_Usabil… ProjectSession5_20251009_Usabil…

STEP 6 — OMNISOUND USABILITY TEST

BEFORE THE TEST

Test plan

Product: Omnisound

Prototype: Hi-fi desktop web prototype

Purpose: Evaluate whether users can understand and use Omnisound’s main functions without receiving instructions about where to click.

Main usability goals being examined:

Goal	What we want to learn
Effectiveness	Can users complete the tasks?
Efficiency	Can users complete them without excessive searching or unnecessary steps?
Learnability	Can a first-time user understand the interface?
Utility	Does the prototype provide the functions users need for these tasks?
Feedback	Does the interface make the result of actions clear?
Visibility	Are important functions and destinations easy to discover?

Target group: Students and similar users who regularly use music, podcast or audiobook applications.

Participants in this simulated example: 3.

Location: Mittuniversitetet, Sundsvall.

Equipment: Laptop with Omnisound running locally in a browser.

Roles:

Person	Role
Yacoub	Test leader
Obid	Observer / notes
Sarshar	Observer / time and clicks

Session 5 explicitly separates the test leader from observers and says the observer records the path, successes, difficulties and signs of misunderstanding. ProjectSession5_20251009_Usabil…

Participant introduction and consent

The test leader reads:

> “Hi. We are students at Mittuniversitetet working on an HCI project called Omnisound.
>
> We are testing the interface, not you, so there are no wrong answers.
>
> Participation is voluntary and you can stop at any time. During the test we will observe how you use the prototype and take anonymous written notes about what happens. The results will only be used for our university project and your name will not appear in our report.
>
> While completing the tasks, please think aloud. Tell us what you are looking for, what you expect to happen, what you find clear and anything that feels confusing.
>
> If you have any questions or feel uncomfortable, please tell us immediately.
>
> Are you comfortable participating?”

Participant response: Yes.

This covers the main consent points in Session 5: voluntary participation, understanding what will happen, observation/possible recording, how results will be used and anonymisation. ProjectSession5_20251009_Usabil…

Pre-test questions

Questions asked to every participant

Question	Purpose
Which audio apps do you normally use?	Understand previous experience
What type of audio do you mainly listen to?	Understand relevant content habits
How often do you use audio applications?	Understand familiarity
Do you normally use mobile, desktop or both?	Understand platform experience

Session 5 says pre-test questions should collect background information relevant to the usability test. ProjectSession5_20251009_Usabil…

Tasks

The participant receives one task at a time.

The course specifically says tasks should describe **what the user should achieve without explaining how to achieve it**. ProjectSession5_20251009_Usabil…

Task 1 — Find and save content

> “You want to listen to Weezer later. Find Weezer and save it somewhere so you can easily find it again.”

Success means:

> Weezer is found → saved → participant can verify it is available in their saved collection.

Task 2 — Resume unfinished audio

> “Earlier you started listening to an audiobook but did not finish it. Find your unfinished audiobook and continue from where you stopped.”

Success means:

> User identifies unfinished audiobook → resumes it → player opens at existing progress.

Task 3 — Browse a particular audio type

> “You are only interested in podcasts right now. Find where you can browse podcasts.”

Success means:

> Participant reaches the Podcasts category.

DURING THE TEST

The test leader uses neutral prompts only when necessary:

> “What are you thinking?”

> “What are you looking for?”

> “What did you expect to happen?”

> “What do you think this control does?”

> “Try what you think makes sense.”

The test leader does not immediately tell the participant which button to press. The exercise says helping too early removes the opportunity to understand why the interface caused difficulty. MDIÖvning20140929

---

p1

Background

Participant: P1

Age: 22

Student: Yes

Main device: iPhone + laptop

Pre-test answers

Which audio apps do you normally use?

> “Mostly Spotify. Sometimes YouTube when I want podcasts.”

What type of audio do you mainly listen to?

> “Music, definitely. Podcasts maybe a couple times a week.”

How often do you use audio apps?

> “Every day.”

Desktop or mobile?

> “Mostly mobile, but Spotify on my computer when I study.”

Task 1 — Find and save Weezer

Participant behaviour

P1 looks at the Home screen for approximately 3 seconds.

Participant says:

> “Okay, I need Weezer. Search seems obvious.”

Clicks **Search** in the left sidebar.

Clicks the search box.

Types:

> Weezer

Weezer appears as the first search result.

Participant says:

> “That was straightforward.”

Clicks Weezer.

Detail page opens.

Participant notices **Play** immediately but pauses before noticing **Save**.

After approximately 2 seconds:

> “Oh, Save is beside Play.”

Clicks **Save**.

Toast confirmation appears.

Participant says:

> “Okay, saved.”

Task leader asks:

> “Do you think the task is complete?”

Participant:

> “Probably, but I'd check Library.”

Clicks **Library**.

Sees Weezer.

> “Yeah, there it is.”

Observer notes

Measure	Result
Completed	Yes
Time	31 sec
Clicks	5
Help	None
Hesitation	Small hesitation finding Save
Wrong actions	None
Important comment	“Search seems obvious.”

Interpretation

Search and Library were easy to discover.

Save feedback was understood.

Save button was slightly less prominent than Play, but participant still found it without help.

Task 2 — Resume unfinished audiobook

Participant returns to Home.

Looks at the hero section.

Scrolls slightly.

Sees **Continue Listening**.

Participant says:

> “I guess this is where unfinished things are.”

Sees The Left Hand of Darkness labelled **Audiobook** and **42% complete**.

Clicks it.

Detail screen shows progress and a **Resume** button.

Participant:

> “Yeah, Resume makes sense.”

Clicks Resume.

Full player opens.

Participant notices progress position.

> “It continued from where it was. That's clear.”

Observer notes

Measure	Result
Completed	Yes
Time	19 sec
Clicks	2
Help	None
Hesitation	None
Wrong actions	None
Important comment	“Resume makes sense.”

Task 3 — Browse podcasts

Participant immediately looks at sidebar.

Clicks **Podcasts**.

Podcast category opens.

> “That was easy.”

Observer notes

Measure	Result
Completed	Yes
Time	4 sec
Clicks	1
Help	None
Hesitation	None

Participant 1 post-test questions

Overall impression?

> “It feels clean. It looks like something I already understand, like Spotify but more organised by audio type.”

What felt easiest?

> “Search and the categories on the left.”

Most confusing?

> “Nothing really confusing. Maybe Save could stand out a bit more.”

Anything expected but couldn't find?

> “Not for those tasks.”

Any confusing labels?

> “No.”

What would you change first?

> “Maybe make Save slightly more obvious.”

Anything important missing?

> “For a real app I'd expect playlists and recommendations, but for this prototype it has the basics.”

p2

Background

Participant: P2

Age: 25

Student: Yes

Main device: Android phone + Windows PC

Pre-test answers

Which audio apps do you normally use?

> “Spotify, Audible and sometimes Pocket Casts.”

Main content?

> “Music and audiobooks.”

How often?

> “Every day.”

Desktop/mobile?

> “Music on both. Audiobooks mainly phone.”

Task 1 — Find and save Weezer

Participant looks at Home.

Instead of clicking Search in the sidebar, participant clicks the large **search field in the top bar**.

Search screen opens.

Participant:

> “I assumed this would search everything.”

Types Weezer.

Result appears.

Clicks Weezer.

Clicks Save quickly.

Toast appears.

Participant:

> “Saved. I assume Library now.”

Clicks Library.

Weezer appears.

Participant:

> “Good.”

Observer notes

Measure	Result
Completed	Yes
Time	24 sec
Clicks	5
Help	None
Hesitation	None
Wrong actions	None
Important observation	Used top search instead of sidebar Search

Interpretation

Having two paths to Search did not confuse the participant.

The participant correctly understood Save → Library.

Task 2 — Resume unfinished audiobook

Participant returns Home.

Looks at sidebar first.

Clicks **Audiobooks**.

Participant scans audiobook cards.

Participant:

> “I know I need an unfinished audiobook, so maybe this page shows one.”

Clicks The Left Hand of Darkness.

Detail screen appears with 42% complete.

Participant:

> “This is probably it.”

Clicks Resume.

Task successfully completed.

Observer notes

Measure	Result
Completed	Yes
Time	27 sec
Clicks	3
Help	None
Hesitation	Moderate
Wrong path	Used Audiobooks instead of Continue Listening
Important comment	“Maybe this page shows one.”

Interpretation

The participant successfully completed the task, but **did not notice/use Continue Listening**.

The interface still allowed the task to be completed through another route.

This may indicate that Continue Listening is not prominent enough for every user.

Task 3 — Browse podcasts

Participant looks at sidebar.

Clicks Podcasts.

Task complete.

Observer notes

Measure	Result
Completed	Yes
Time	5 sec
Clicks	1
Help	None

Participant 2 post-test questions

Overall impression?

> “Pretty intuitive. I like having the categories separated.”

Easiest?

> “Search and Library.”

Most confusing?

> “I didn't immediately notice Continue Listening. I went to Audiobooks instead.”

Anything missing?

> “Maybe some stronger indicator that unfinished things are on the home page.”

Confusing labels?

> “No.”

What would you change first?

> “Make Continue Listening more visible or move it closer to the top.”

Anything else?

> “I'd probably use the left navigation a lot. It feels predictable.”

p3

Background

Participant: P3

Age: 20

Student: Yes

Main device: Android phone

Pre-test answers

Which audio apps do you normally use?

> “YouTube Music and Spotify.”

What do you mainly listen to?

> “Music. Very rarely podcasts.”

How often?

> “Daily.”

Desktop/mobile?

> “Almost always phone.”

Task 1 — Find and save Weezer

Participant looks at Home.

Clicks **Music** first.

Looks through cards.

Sees Weezer already visible in Music.

Clicks Weezer.

Detail screen appears.

Participant clicks the heart/save control.

Participant says:

> “I assume this adds it somewhere.”

Toast confirms:

> Weezer saved to your Library.

Participant:

> “Okay, Library.”

Clicks Library.

Weezer visible.

Observer notes

Measure	Result
Completed	Yes
Time	28 sec
Clicks	4
Help	None

Measure	Result
Hesitation	Small
Alternative route	Music instead of Search
Important comment	“I assume this adds it somewhere.”

Interpretation

Task was completed successfully without Search.

The prototype supports more than one valid route.

Feedback from the save toast helped confirm what happened.

Task 2 — Resume unfinished audiobook

Participant returns Home.

Looks at hero text.

Scrolls past Continue Listening without clicking.

Clicks Library.

Participant says:

> “Maybe unfinished things are saved here.”

Sees The Left Hand of Darkness.

Clicks it.

Detail page shows 42% complete and Resume.

Participant:

> “Oh yeah, that's obviously unfinished.”

Clicks Resume.

Full player opens.

Observer notes

Measure	Result
Completed	Yes
Time	35 sec
Clicks	4
Help	None
Hesitation	Moderate
Wrong/alternative route	Library rather than Continue Listening
Important comment	“Maybe unfinished things are saved here.”

Interpretation

Again, Continue Listening was not the participant's first choice.

Library successfully supported recovery, so the task remained effective.

Efficiency may be weaker because the most direct route was overlooked.

Task 3 — Browse podcasts

Participant looks at sidebar.

Clicks Music by mistake.

Immediately notices Music is highlighted.

Participant says:

> “Oops, wrong one.”

Clicks Podcasts.

Observer notes

Measure	Result
Completed	Yes
Time	7 sec
Clicks	2
Help	None
Error	Clicked neighbouring category accidentally
Recovery	Immediate
Important observation	Active navigation highlight made error visible

Interpretation

This is a small user error rather than a major usability problem.

Feedback from the active navigation state made recovery immediate.

Participant 3 post-test questions

Overall impression?

> “It's easy to understand. The sidebar does most of the work.”

Easiest?

> “Finding categories.”

Most confusing?

> “I didn't really notice the Continue Listening section.”

Anything expected but couldn't find?

> “No.”

Confusing labels?

> “No.”

What would you change?

> “Maybe make Continue Listening more obvious. The big banner above it takes a lot of space.”

Anything missing?

> “Maybe recently played, but that's not really necessary for these tasks.”

AFTER THE TEST

Combined results

Task completion

Task	P1	P2	P3
Find and save Weezer	Success	Success	Success
Resume unfinished audiobook	Success	Success	Success
Browse podcasts	Success	Success	Success

Overall effectiveness

9/9 tasks completed successfully.

No participant required direct help from the test leader.

Efficiency overview

Task	P1	P2	P3
Search/save	31 sec	24 sec	28 sec
Resume audiobook	19 sec	27 sec	35 sec
Browse podcasts	4 sec	5 sec	7 sec

The largest difference occurred during the **resume audiobook task**.

P1 used:

> Home → Continue Listening → Audiobook → Resume

P2 used:

> Home → Audiobooks → Audiobook → Resume

P3 used:

> Home → Library → Audiobook → Resume

All routes worked, but only one participant used the shortest intended route.

Main findings

Finding 1 — Continue Listening is easy to overlook

Evidence:

P1 found it immediately.

P2 instead used Audiobooks.

P3 instead used Library.

Two of three participants did not use the intended Continue Listening route.

Possible HCI connection:

Visibility.

The feature exists and works, but its visual prominence may be weaker than surrounding content.

Severity: Medium–High.

It does not prevent users from completing the task, but it reduces Efficiency.

Finding 2 — Search is highly understandable

P1 used sidebar Search.

P2 used top Search.

P3 found Weezer through Music instead.

All participants successfully located the requested content.

Possible HCI connection:

Visibility + Effectiveness.

Multiple routes allowed users to rely on their own mental model.

Severity: No problem.

Decision: Keep.

Finding 3 — Saving content works well

All three participants correctly understood the save action.

The toast:

> “Weezer saved to your Library”

made the consequence explicit.

All participants could then find the item in Library.

Possible HCI connection:

Feedback.

Severity: No problem.

P1 suggested making the Save control slightly more visually prominent.

Decision: Minor visual improvement only.

Finding 4 — Content categories are easy to understand

All participants understood Music, Podcasts and Audiobooks.

P3 accidentally clicked Music once but immediately recovered after seeing the active highlight.

Possible HCI connection:

Consistency + Feedback + Visibility.

Severity: Low / no redesign required.

Finding 5 — Library acts as a useful recovery path

P3 used Library instead of Continue Listening and still successfully found the audiobook.

Participants understood that saved items belonged there.

Possible HCI connection:

Consistency + mental model + Utility.

Decision: Keep current Library structure.

Prioritized usability problems

Priority	Problem	Evidence	Impact
1 — Medium/ High	Continue Listening is not prominent enough	2/3 participants ignored it	Longer route to unfinished content
2 — Low	Save button could be slightly more prominent	P1 hesitated briefly	Small Learnability/Efficiency issue
3 — Low	Large Home hero uses substantial space	P3 mentioned it	May push useful content downward
No issue	Search	All users succeeded	Keep
No issue	Library	All users understood it	Keep
No issue	Category navigation	All completed task	Keep

Proposed redesign

Change 1 — Make Continue Listening more prominent

Current issue

The large introductory hero occupies much of the top of Home.

Continue Listening begins underneath it.

Two participants looked elsewhere for unfinished content.

Redesign

Reduce the height of the hero area.

Move **Continue Listening** closer to the top.

Possibly change:

> CONTINUE LISTENING
>
> Back in progress

to something more direct:

> **Continue Listening**

and increase contrast/size slightly.

Keep progress percentages visible.

Goal

Improve:

Visibility → Efficiency

without changing the basic navigation.

Change 2 — Improve Save prominence slightly

Current issue

One participant needed a short moment to notice Save next to Play.

Redesign

Keep Save in the same position but increase its visual clarity slightly.

For example:

> ♡ Save

and after clicking:

> ♥ Saved

Keep the toast confirmation.

Goal

Improve:

Affordance + Feedback

without making Save compete visually with Play.

Change 3 — Reduce unnecessary hero space

The banner:

> “Pick up where your day left off.”

looks visually polished but takes substantial vertical space.

Reducing it slightly would let the user's actual content appear sooner.

This fits the usability exercise's principle that irrelevant or lower-priority information should not compete visually with the important task content. MDIÖvning20140929

Things we would NOT redesign

We would keep:

Sidebar navigation

because all participants understood it.

Separate Music / Podcasts / Audiobooks categories

because they were easy to recognise.

Search

because participants found it using different valid routes.

Library

because participants correctly understood it as saved content.

Save confirmation toast

because it clearly communicated the action result.

Playback Resume wording

because users immediately understood what it meant.

This is important: usability testing does **not** mean changing everything. Features that worked should remain stable.

Example action list

The usability exercise says testing should result in a prioritised action list and redesign proposals. MDIÖvning20140929

Action	Priority	Decision
Reduce Home hero height	High	Implement
Move Continue Listening upward	High	Implement
Increase Continue Listening visual prominence	High	Implement
Slightly strengthen Save button appearance	Low	Implement if time allows
Redesign Search	None	Do not change
Redesign Library	None	Do not change
Change sidebar categories	None	Do not change

Hi-fi version 2

After implementing the changes, save a new prototype version rather than overwriting all evidence of the old one.

Your project history becomes:

> **Lo-fi sketches**
>
> ↓
>
> **Hi-fi Version 1**
>
> ↓
>
> **Usability Test**
>
> ↓
>
> Finding: Continue Listening overlooked
>
> ↓
>
> **Redesign**
>
> ↓
>
> **Hi-fi Version 2**

That is exactly the kind of evidence you need for the iterative process later.

Example report-ready summary

Later, your report could describe this example test like this:

> “A think-aloud usability test was conducted with three participants familiar with digital audio services. Participants completed three tasks involving finding and saving music, resuming unfinished audio and browsing a specific content category. All participants completed all tasks without direct assistance.
>
> Search, Library and category navigation were generally understood immediately. However, two of the three participants did not use the Continue Listening section when asked to resume an unfinished audiobook. Instead, they navigated through Audiobooks or Library. Although both alternative routes were successful, this indicated that the Continue Listening feature had weaker visibility than intended.
>
> Based on the findings, the Home page was redesigned by reducing the size of the introductory hero area and increasing the prominence of Continue Listening. The Save action was also made slightly more visible while retaining its confirmation feedback.”

That is the type of final result you should expect from Step 6.

Complete simulated Step 6 package

You now have an example of:

test plan → participant introduction/consent → pre-test questions → realistic tasks → three complete usability sessions → observation notes → task measurements → post-test interviews → combined results → usability problems → severity priorities → redesign recommendations → iteration plan

For the real project, the structure can stay almost exactly like this. **Only the participant behaviour, quotes, measurements, findings and resulting redesign decisions need to come from your actual tests.**

OMNISOUND · HCI PROJECT · GROUP 1

V2 Usability Testing

Step 7 · Second usability evaluation · English-language Notion notes

> **Source note:** This PDF is a formatted conversion of the supplied Notion text. It preserves its statements identifying participant accounts and measurements as **simulated examples**. The PDF is not independent verification of participant testing.

- v2 code

omnisound-v2.zip

- Run v2 prototype

file:///C:/Users/Jake_/Desktop/ui%20proj/omnisound-v2/index.html

Exactly — **Step 7 is V1 → findings → V2 → test V2 again**. P3 requires the design process to be iterative, and a single iteration is explicitly not acceptable. P3 Projekt (2)

For the second test, you **should not make everything completely different**, because then you cannot tell whether your redesign fixed the original problems. We keep the same core goals, use **new participants**, and focus especially on the V2 changes: **Continue Listening visibility, Save/Saved clarity, and the smaller Home hero**.

Everything below is **simulated example data**, not real participant results.

BEFORE THE TEST V2

Test plan

Product: Omnisound

Prototype: Hi-fi Version 2

Purpose: Evaluate whether the redesigns made after the first usability test improved the identified usability problems.

The first test suggested:

- Continue Listening was too easy to overlook.
- The Home hero occupied too much visual space.
- Save was understood, but could be more noticeable.

V2 therefore changed:

> smaller Home hero
>
> → more prominent Continue Listening
>
> → clearer ♡ Save / ♥ Saved state

The second usability test investigates whether these changes actually helped.

Main usability goals

Goal	What we want to learn
Effectiveness	Can users complete the tasks?

Goal	What we want to learn
Efficiency	Can they complete them without unnecessary searching?
Learnability	Can first-time users understand V2 quickly?
Visibility	Is Continue Listening easier to discover?
Affordance	Is Save clearly recognisable as an action?
Feedback	Is it obvious when content has been saved?

Target group

Students and similar users who regularly use music, podcast or audiobook applications.

Participants

For this simulated example:

3 new participants: P4, P5 and P6.

Using new participants is useful because they have not already learned the interface from Test 1.

Equipment

Laptop running Omnisound V2 in a browser.

Roles

Person	Role
Yacoub	Test leader
Obid	Observer / notes
Sarshar	Observer / time and clicks

Session 5 says the test leader leads while the observer records routes, successes, difficulties and signs that the participant does not understand something. ProjectSession5_20251009_Usabil...

Participant introduction / consent

The test leader says:

> “Hi. We are students at Mittuniversitetet evaluating a prototype called Omnisound.
>
> We are testing the interface, not you, so there are no wrong answers.
>
> Participation is voluntary and you can stop at any time. We will observe how you use the prototype and take anonymous written notes. The information will only be used for our university project.
>
> Please think aloud while completing the tasks. Tell us what you are looking for, what you expect to happen and anything that feels unclear.
>
> If you have any questions or feel uncomfortable, please tell us.
>
> Are you comfortable participating?”

Participant response: Yes.

The course requires participants to understand that participation is voluntary, that interaction may be observed, how results will be used, and that they will be anonymised. ProjectSession5_20251009_Usabil…

Pre-test questions

Ask each participant:

> “Which audio applications do you normally use?”

> “Do you mainly listen to music, podcasts, audiobooks, or a mixture?”

> “How often do you use audio applications?”

> “Do you mainly use them on mobile, desktop, or both?”

Session 5 recommends collecting relevant background information before testing. ProjectSession5_20251009_Usabil…

Tasks

The tasks still describe **what the participant should achieve, not which controls to use**, exactly as Session 5 requires. ProjectSession5_20251009_Usabil…

Task 1 — Find and save content

> “You want to listen to Weezer later. Find Weezer and save it so you can easily find it again.”

This specifically checks whether the redesigned ♡ **Save / ♥ Saved** control is clearer.

Success:

> Weezer found → saved → participant understands that it has been saved → can find it in Library.

Task 2 — Continue unfinished content

> “Earlier you started listening to an audiobook but did not finish it. Find your unfinished audiobook and continue listening from where you stopped.”

This is deliberately similar to Test 1 because **Continue Listening was the biggest V1 problem**.

Success:

> participant notices unfinished content → selects audiobook → Resume → player opens at existing progress.

Task 3 — Browse podcasts

> “You want to look only at podcasts. Find the part of Omnisound where you can browse podcasts.”

This checks that the redesign did not accidentally harm navigation that already worked well.

Success:

> Podcasts category reached.

DURING THE TEST V2

Participants are asked to think aloud.

The test leader may use:

> “What are you thinking?”

> “What are you looking for?”

> “What did you expect to happen?”

> “What do you think this control does?”

If asked:

> “Should I click this?”

respond:

> “Try what you think makes sense.”

The course specifically recommends letting the participant work things out rather than immediately helping, because their behaviour and reactions are the actual usability data. ProjectSession5_20251009_Usabil...

SIMULATED PARTICIPANT 4

Background

Participant: P4

Age: 23

Student: Yes

Main devices: Windows laptop + Android phone

Pre-test answers

Which apps do you use?

> “Mostly Spotify and YouTube Music.”

Main audio type?

> “Mostly music, but sometimes podcasts.”

How often?

> “Every day.”

Platform?

> “Phone normally. Computer when studying.”

Task 1 — Find and save Weezer

P4 looks at Home for approximately 2 seconds.

Clicks the top search field.

Types:

> Weezer

Opens Weezer.

On the detail page, P4 immediately looks at **Play** and ♡ **Save**.

Participant:

> “Save is pretty obvious.”

Clicks Save.

Button changes:

> ♥ Saved

Toast appears:

> “Weezer saved to your Library.”

Participant says:

> “Okay, that's definitely saved.”

Clicks Library.

Finds Weezer.

Observer notes

Measure	Result
Completed	Yes
Time	22 sec
Clicks	5
Help	None
Hesitation	None
Wrong actions	None
Comment	“Save is pretty obvious.”

Interpretation

The redesigned Save control was noticed immediately.

Both the changed button state and toast gave feedback confirming the action.

Task 2 — Resume unfinished audiobook

P4 returns Home.

Immediately scans below the hero.

Participant:

> “Continue Listening. Probably there.”

Clicks **The Left Hand of Darkness**.

Sees existing progress and Resume.

Clicks Resume.

Player opens at the existing position.

Participant:

> “Yeah, that's exactly what I'd expect.”

Observer notes

Measure	Result
Completed	Yes
Time	13 sec
Clicks	2
Help	None
Hesitation	None
Route	Home → Continue Listening → audiobook → Resume
Comment	“Probably there.”

Task 3 — Browse podcasts

P4 immediately clicks Podcasts in the sidebar.

Observer notes

Measure	Result
Completed	Yes
Time	4 sec
Clicks	1
Help	None

Participant 4 post-test

Overall impression?

> “Very straightforward. It looks familiar without being exactly Spotify.”

Easiest part?

> “The sidebar and searching.”

Most confusing?

> “Nothing major.”

Did anything feel difficult to discover?

> “No, Continue Listening was pretty visible.”

What would you change?

> “Maybe nothing for these tasks.”

Was Save clear?

> “Yes. The heart and word Save made it obvious.”

SIMULATED PARTICIPANT 5

Background

Participant: P5

Age: 26

Student: Yes

Main devices: iPhone + MacBook

Pre-test answers

Apps?

> “Spotify, Apple Podcasts and Storytel.”

Audio type?

> “Music and audiobooks mostly.”

Frequency?

> “Every day.”

Platform?

> “Mostly phone.”

Task 1 — Find and save Weezer

P5 clicks Search in sidebar.

Searches Weezer.

Opens result.

Participant briefly looks between **Play** and **Save**.

Approximately one-second pause.

> “I'll save this.”

Clicks Save.

Button visibly changes to Saved.

Participant:

> “Okay, now I know it worked.”

Clicks Library and confirms Weezer appears.

Observer notes

Measure	Result
Completed	Yes
Time	24 sec
Clicks	5
Help	None
Hesitation	Very small
Wrong actions	None
Comment	“Now I know it worked.”

Interpretation

No meaningful usability problem remained.

The saved state clearly communicated feedback.

Task 2 — Resume unfinished audiobook

P5 reaches Home.

Sees:

> **Continue Listening**

without scrolling.

Participant:

> “This looks like what you're asking for.”

Clicks audiobook.

Clicks Resume.

Player opens.

Participant checks progress.

> “It's already partway through, so yes.”

Observer notes

Measure	Result
Completed	Yes
Time	15 sec
Clicks	2
Help	None
Hesitation	None
Route	Continue Listening directly

Task 3 — Browse podcasts

Participant clicks Podcasts.

Observer notes

Measure	Result
Completed	Yes
Time	4 sec
Clicks	1
Help	None

Participant 5 post-test

Overall impression?

> “It feels organised. I understand the separation between music, podcasts and books.”

Easiest?

> “Continue Listening and navigation.”

Most difficult?

> “Nothing really.”

Anything unclear?

> “No.”

Save button?

> “Clear.”

What would you change?

> “Maybe I'd want Recently Played eventually, but that's another feature.”

This is **not automatically a redesign requirement** because it is outside the core tasks being tested.

SIMULATED PARTICIPANT 6

Background

Participant: P6

Age: 21

Student: Yes

Main device: Android phone + Windows laptop

Pre-test answers

Apps?

> “Spotify and Audible.”

Content?

> “Mostly music and audiobooks.”

How often?

> “Probably a few hours every day.”

Platform?

> “Phone mostly, but laptop too.”

Task 1 — Find and save Weezer

P6 initially clicks **Music** rather than Search.

Finds Weezer directly.

Participant:

> “It's already here, so I don't need Search.”

Opens Weezer.

Immediately clicks Save.

Button becomes Saved and toast appears.

> “Good.”

Clicks Library.

Weezer visible.

Observer notes

Measure	Result
Completed	Yes
Time	21 sec
Clicks	4
Help	None
Hesitation	None
Alternative route	Music rather than Search
Comment	“I don't need Search.”

Interpretation

Alternative navigation remained effective.

The redesigned Save control worked regardless of the route used to find the content.

Task 2 — Resume unfinished audiobook

Participant returns Home.

Reads:

> Continue Listening.

Immediately clicks The Left Hand of Darkness.

Participant:

> “That's definitely unfinished because it shows 42%.”

Clicks Resume.

Player opens.

Observer notes

Measure	Result
Completed	Yes
Time	12 sec
Clicks	2
Help	None

Measure	Result
Hesitation	None
Comment	“It shows 42%.”

Interpretation

Both the heading and progress information supported recognition.

Task 3 — Browse podcasts

P6 clicks Podcasts.

Task completed immediately.

Observer notes

Measure	Result
Completed	Yes
Time	3 sec
Clicks	1
Help	None

Participant 6 post-test

Overall impression?

> “Easy. I basically knew where everything was without being told.”

Easiest?

> “Categories and Continue Listening.”

Most confusing?

> “Nothing in these tasks.”

Any unclear labels?

> “No.”

What would you change?

> “Maybe let me sort the Library, but that's not really necessary here.”

Was Save understandable?

> “Yes.”

AFTER THE TEST V2

Session 5 says the post-test phase should gather overall impressions, what worked, what did not, possible improvements and missing elements, followed by analysis and redesign. ProjectSession5_20251009_Usabil...

Combined results

Task completion

Task	P4	P5	P6
Find/save Weezer	Success	Success	Success
Resume unfinished audiobook	Success	Success	Success
Browse podcasts	Success	Success	Success

Effectiveness

9/9 tasks completed.

No participant required help.

Same completion rate as V1, so V1 was already effective.

The important question is whether V2 improved **efficiency and discoverability**.

Efficiency

Task	V1 range	V2 P4	V2 P5	V2 P6
Find/save	24–31 sec	22 sec	24 sec	21 sec
Resume	19–35 sec	13 sec	15 sec	12 sec
Browse Podcasts	4–7 sec	4 sec	4 sec	3 sec

The largest improvement occurred in the **Resume task**, which is exactly where V1 showed the largest usability problem.

Compare the Continue Listening behaviour

V1

Only P1 used:

> Home → Continue Listening → audiobook → Resume

P2 used:

> Home → Audiobooks → audiobook → Resume

P3 used:

> Home → Library → audiobook → Resume

Therefore:

> **1/3 used Continue Listening directly.**

V2

P4:

> Home → Continue Listening → audiobook → Resume

P5:

> Home → Continue Listening → audiobook → Resume

P6:

> Home → Continue Listening → audiobook → Resume

Therefore:

> **3/3 used Continue Listening directly.**

In this simulated example, that is strong evidence that the redesign addressed the original Visibility problem.

Main finding 1 — Continue Listening improved

V1 problem

Two of three participants overlooked it.

V2 redesign

- smaller hero
- Continue Listening moved visually closer to top
- stronger heading
- lime marker
- progress information retained

V2 result

All three participants noticed and used it immediately.

HCI interpretation

> **Visibility → Efficiency**

Making the function easier to discover reduced unnecessary navigation.

Status

Original problem appears reduced.

Finding 2 — Save/Saved state improved

V1

One participant briefly struggled to notice Save.

V2

All three participants recognised the action.

After clicking it, the combination of:

> ♥ Saved

and the toast message clearly confirmed the result.

HCI interpretation

> **Affordance + Feedback**

The control suggests what can be done, and the resulting state communicates what happened.

Status

No significant issue observed in V2.

Finding 3 — Smaller hero helped prioritise content

None of the V2 participants commented that the hero blocked useful information.

Continue Listening appeared sooner and was visible without needing users to search further down the page.

HCI interpretation

The important task-related information competes less with decorative content.

Status

Keep V2 design.

Finding 4 — Existing navigation still works

All participants successfully used:

- Music
- Podcasts
- Library
- Search

No navigation regression appeared after redesigning Home.

Status

No change required.

New observations from V2

The second usability test may still produce new ideas.

For example:

P5 mentioned:

> Recently Played.

P6 mentioned:

> Library sorting.

These may be useful future ideas, but they are **not automatically usability problems**.

Neither prevented the tested tasks from being completed.

Therefore:

> do not expand the project scope just because somebody suggested another feature.

Prioritised V2 problems

Priority	Finding	Evidence	Decision
No major issue	Continue Listening	3/3 found directly	Keep V2
No major issue	Save/Saved	3/3 understood	Keep V2
No major issue	Search	Worked for relevant participants	Keep
No major issue	Categories	3/3 successful	Keep
Low / future	Recently Played	One suggestion	Outside current scope
Low / future	Library sorting	One suggestion	Outside current scope

Redesign suggestions after Test 2

Here the correct answer can genuinely be:

> **No major redesign is required for the tested workflows.**

Usability testing does not require you to invent another problem.

The current design should therefore keep:

- compact hero
- prominent Continue Listening
- progress information
- ♡ Save / ♥ Saved
- save confirmation toast
- sidebar
- Library

- Search
- separate content categories

Potential future improvements such as Recently Played or Library sorting can be documented as **future work**, rather than implemented now.

Final V1 → V2 iteration table

This is probably the single most useful table for your report:

V1 finding	V2 change	HCI justification	Test 2 result
Continue Listening overlooked by 2/3	Reduced hero and made section more prominent	Visibility / Efficiency	3/3 used it directly
Resume route sometimes inefficient	Important unfinished content moved higher	Efficiency	Resume times reduced
Save slightly easy to overlook	Clearer ♡ Save / ♥ Saved styling	Affordance	3/3 recognised action
Result of Save needed to remain clear	Retained toast + Saved state	Feedback	Users understood result
Search worked well	No redesign	Preserve successful design	Still worked
Library worked well	No redesign	Preserve successful design	Still worked
Category navigation worked	No redesign	Consistency	Still worked

Step 7 conclusion

A report-ready version could later say:

> “Following the first usability evaluation, a second version of the Omnisound prototype was produced. The redesign focused specifically on the problems identified in the first iteration rather than changing areas that had already performed successfully.
>
> The introductory Home hero was reduced and the Continue Listening section was given greater visual prominence. The Save action was also redesigned with clearer Save and Saved states while retaining confirmation feedback.
>
> A second think-aloud usability test was then conducted using the same core tasks with new participants. All participants completed all three tasks without assistance.
>
> In the first test, only one of three participants used Continue Listening directly when asked to resume unfinished content. In the second test, all three participants used the section directly. The simulated results therefore suggest that the change improved the visibility and efficiency of this workflow.
>
> No major usability problems were observed in the tested V2 workflows, so the successful parts of the design were retained rather than introducing unnecessary further changes.”

Step 7 is finished when you have

> **Hi-fi V1**
>
> ↓
>
> **Test 1**
>
> ↓
>
> **documented findings**
>
> ↓
>
> **justified redesign**
>
> ↓
>
> **Hi-fi V2**
>
> ↓
>
> **Test 2**
>
> ↓
>
> **comparison of V1 and V2**

That gives you the traceable iterative design process P3 is asking for, rather than simply saying *“we made a second prototype.”*